

WEB CLIENT APPLICATION

User Manual

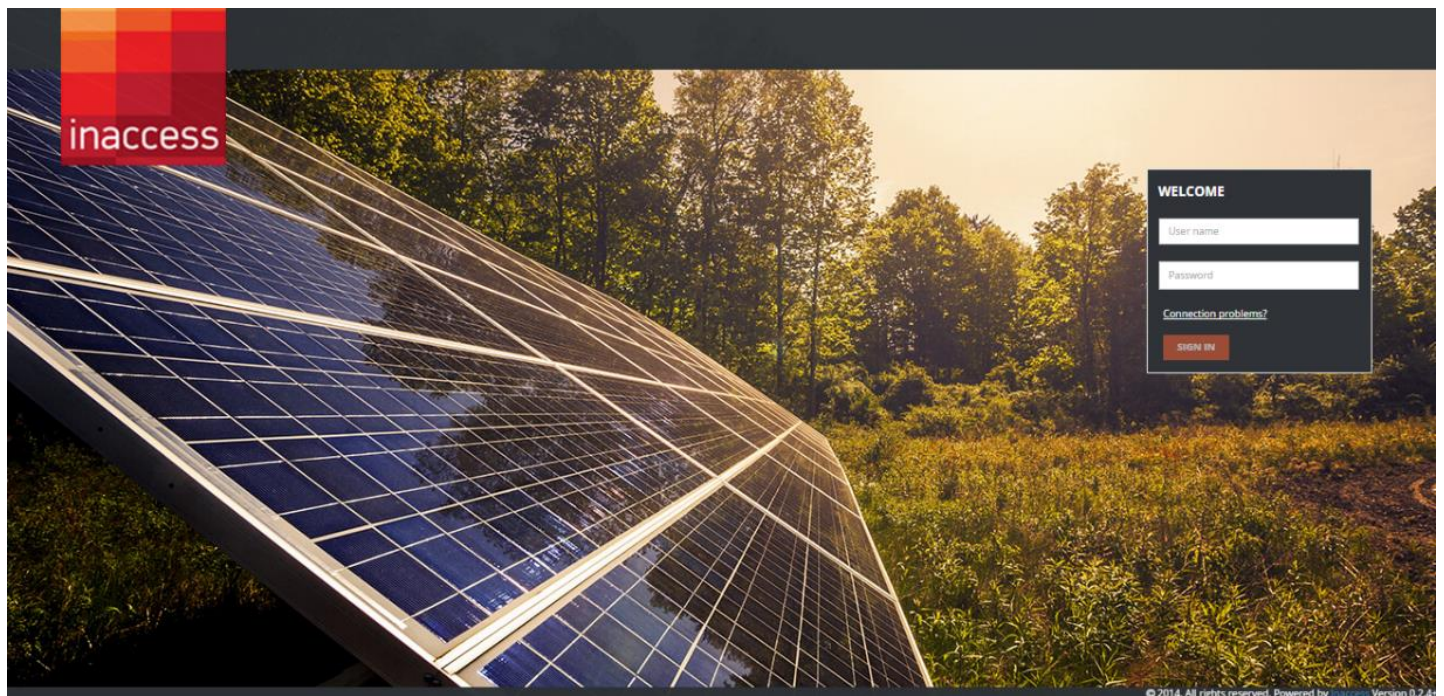




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System requirements

➤ The system requirements for the solution Web client are:
Browser: Internet Explorer v8.0 and above, Mozilla Firefox v3.0 and above, Google Chrome v5.0 and above, Safari 8 and above.

Note: The Inaccess solution is a Web client application based on Javascript technology. The different browsers available on the market have various Javascript support performances: Google Chrome currently provides the best user experience.

Most video surveillance products offer an ActiveX plugin so that they can be viewed/operated over the Web. ActiveX works only within Internet Explorer, so in certain cases, the video surveillance cannot be operated in Google Chrome or Mozilla Firefox.



The login page

In order for you to access the in Inaccess web application, insert the appropriate credentials (username/password) at the login page and press the “Login” button.

The URL is <https://portal-new.solarpark-online.com/>

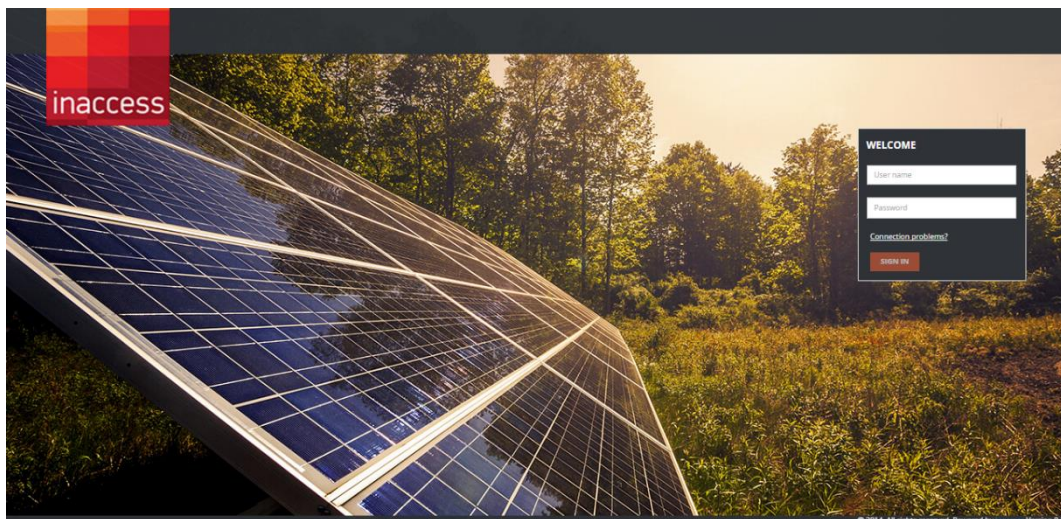


Figure1.

The Inaccess platform login page

If you cannot login, a red exclamation mark appears on the top of the “Welcome” frame with a short description of the failure cause.

Important Notice: In cases where logging failure persists, you should consult the Inaccess support team through the link “Connection Problems?”

The login and password are case sensitive.



The application page

After successfully logging into the system, you will enter the application environment that is composed of the following elements:

- An auto- hide navigation panel
- A header panel
- A content area

The navigation panel

The navigation panel is a popup menu located on the top-left side of the application page and it contains the various buttons required for an efficient User navigation. It appears whenever you click on the

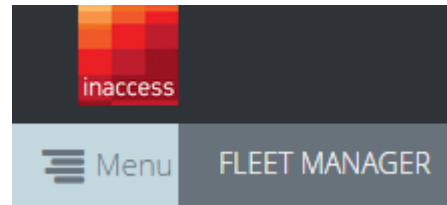


Figure2.
The navigation
popup-menu
activation
button

Figure 3. The navigation menu without any active PV Plant site selection

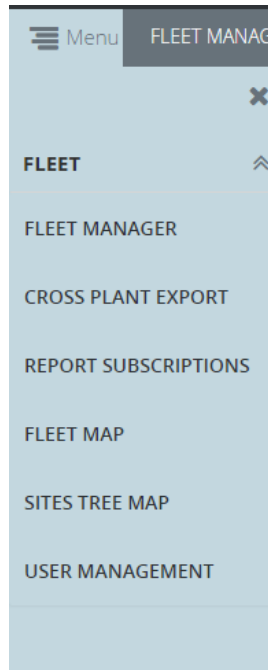
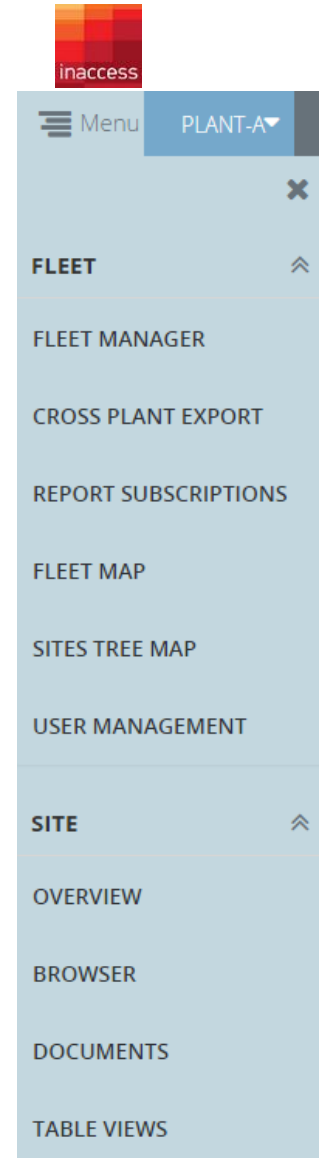


Figure 4. The navigation menu with a selected PV Plant site

As it can be easily understood the menu has now been automatically reformed with the addition of an extra command subset relevant to site management functional activities named Site.



The header panel

The header panel is the topmost panel of the application frame. It has two discrete groups of commands.

At the right, in the back bar, is a permanent command set with three buttons:

- user profile button (a user photo is shown when available),
- language switching button
- content refresh button.

At the left in the grey bar is a command set which is formed depending on whether the application is looking at the fleet level or at a single PV Plant. The first item is the navigation control button. The drop down menu which this deploys also depends on whether the application is looking at the fleet or single plant level. At the fleet level, this is the only active control.

Figure 5. The initial form of header panel



Next to this is the configuration button, which is active when a plant is selected. In this case, the button can be used to switch between PV Plants. To the right of this is the Time-span control group. This set of buttons offers various options to define the desired time period.

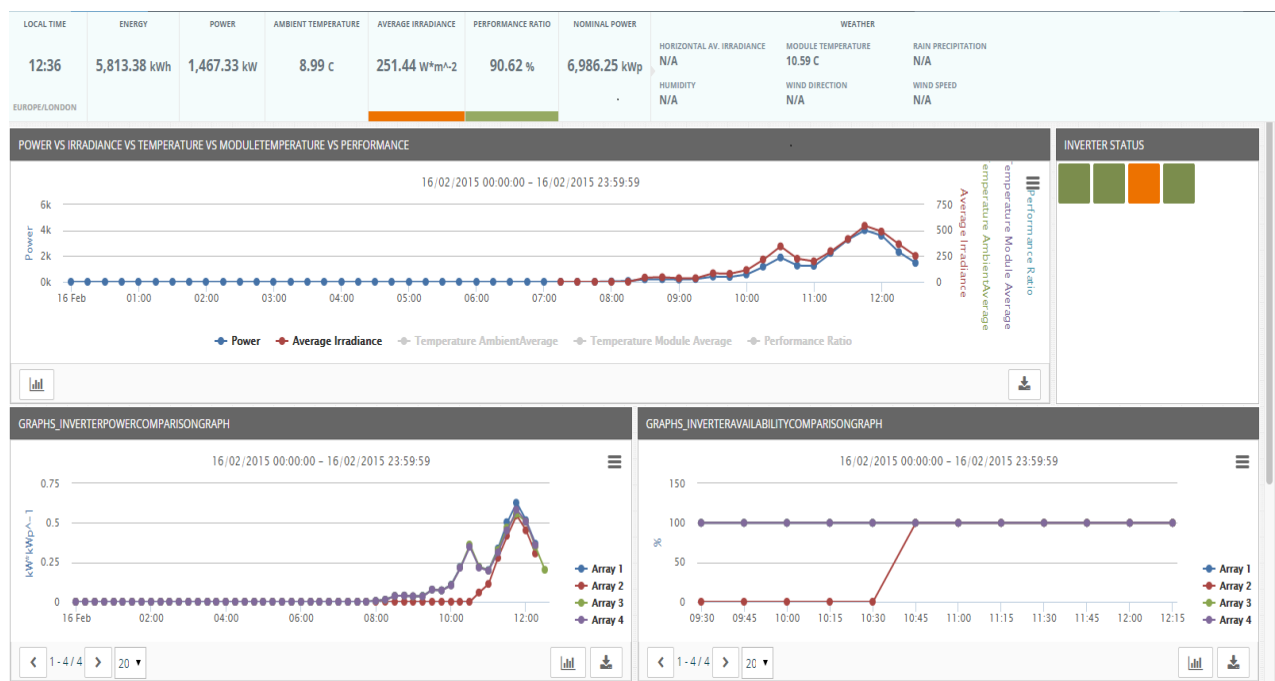
Figure 6. The header panel reformation after the proper PV Plant site selection



The content area

The content area is below the header panel, and occupies the main area of the screen. This is the main application frame and is dictated by your command selections made in the navigation control panel which uses the left section of the frame. We will return to a discussion of the content area after exploring the commands which control its content.

Figure 7. The multiple forms of Inaccess platform content area



Using the Header Panel

Header Panel - User Profile Configuration

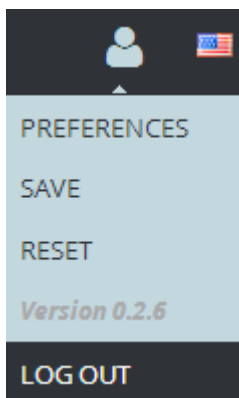


Figure 8. The User Profile popup menu

From this menu, the user can change the information associated with their user account under PREFERENCES, save the changes using the SAVE command, return the configuration of the overall display to the original default under RESET, and LOGOUT.

PREFERENCES: by selecting this command, the system activates the PREFERENCES configuration dialog. This administrative dialog contains two individual information segments named “User Information” and “User Parameters.” These contain both mandatory and optional information fields which enable the system to properly distinguish users and deliver the features of their corresponding roles.

User Information Configuration

Figure 9.

Your Information Configuration section

User Information

User gstefan	First Name	Mobile () - ____
Password *****	Surname	Phone Number () - ____
Confirm *****	Email gstefan@inaccess.com	

Save

This section is displayed on the top of PREFERENCES dialog and gives you the ability to configure the following parameters. (Any grayed out fields are not changeable and are exclusively defined by the system administrator during user profile account creation phase).

- **User:** the login name used in the log in process. You will see this name displayed whenever you locate the mouse cursor on top of user profile icon in the header panel. This parameter is given by system administrator and is unchangeable for the life of the user profile.
- **Password:** the user’s choice of password to use in the log in process. The password may be reset at any time by the user, or reset by the administrator. There are no restrictions on content or word length.
- **Confirm (Password):** Mandatory information that assures the proper definition of your new password string.
- **First Name:** the user’s first name is optional
- **Surname:** the user’s last name is optional
- **Mobile (Number):** a cellular phone number can be provided to enable SMS alerting services.



- **Phone Number:** a second phone number may be provided for administrative use
- **Email:** and email address is required in order to receive alerts or reports

You have the ability to amend any of the above informational fields (apart from the User name) and permanently store any change on system's database by pressing the "SAVE" button on the bottom of this section

User Parameters Configuration

Figure10. The User Parameters Configuration section

You have the ability to define formatting criteria for certain system information. This section is positioned in the middle of the PREFERENCES dialog and gives the ability to configure the following parameters:

- **Date format:** select the date display format that system will use in the content of the Web application and within any exported report and forwarded notification context.

You have the ability to amend and permanently store any change on the above parameters by pressing the "SAVE" button on the bottom of this section

User Parameters

CMMS Parameters Configuration

The inaccess web based platform also provides you with the ability to associate your profile with advanced PV Plant **maintenance activities** supported by a fully dedicated Computerized Maintenance Management System (CMMS). This CMMS system can interoperate with the Inaccess platform in order to apply maintenance management functionality.

These parameters must be defined in the CMMS Information section.

Figure11. The Computerized Maintenance Management System Configuration Parameters

CMMS Information

In order to activate the provisioning of maintenance management services you have to specify the following parameters:

- Base URI, identifying the CMMS server
- The Application Key

- The Authorization Token
- Private Key

The three keys refer to an API instance defined within CMMS. A CMMS service administrator must provide these keys in order to be applied in CMMS Information section by the User.

Using the Navigation Panel

The navigation panel is a popup menu that offers quick access to several monitoring and management system modules. The Navigation panel offers varying features from general inspection to advanced monitoring, notification and reporting services, both for individual PV Plant sites or groups of PV Plant sites. Submenus appear as needed. Simply click on the paragraph icon in the header panel to open the menu. Refer to figure 5.

The Fleet Manager page

Fleet Manager is the default display immediately after the login screen, and is also first option in the Navigation Panel. It delivers an overview of the plants that are authorized to appear in your portfolio. As this part of the “FLEET” subsection of the menu, it is active when no individual plant has been selected. To return to this view from any display in the application, simply select FLEET MANAGER in the Navigation Panel. Figure 12. Using the “Fleet Manager” command button

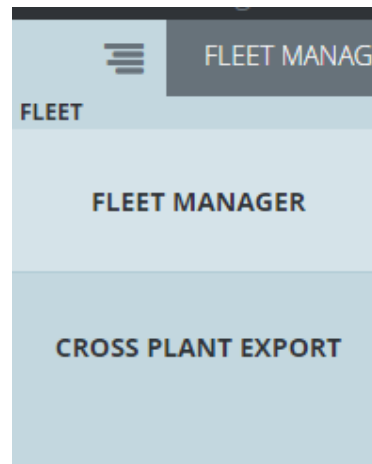
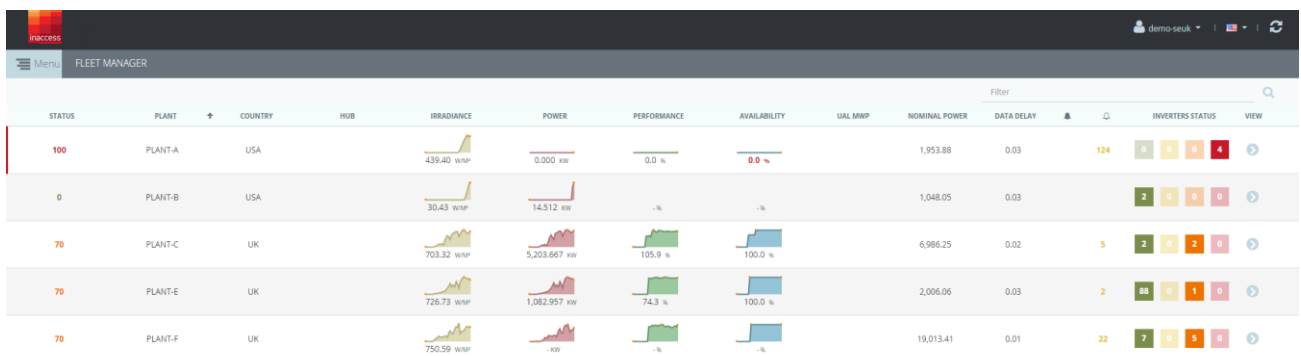


Figure 13. The Fleet Manager page



Fleet Management Information Set

Fleet Manager displays a list of periodically updated information for your PV Plants. (it is possible to force an update with the chasing arrows icon to the right of the language flag.) The information categories are explained below.

- **NAME:** the name of the PV Plant
- **COUNTRY:** the country in which PV Plant is located
- **HUB:** the parental hub node (or subgroup, if any) to which the PV Plant belongs
- **IRRADIANCE:** the average irradiance (in W/M²) taken over all sensors at the plant
- **POWER:** the current production (in kW) for the entire plant
- **PERFORMANCE:** the calculated performance ratio (as a percentage) of the PV Plant
- **AVAILABILITY:** the operational availability of the plant's assets (as a percentage)
- **UAL MWP:** The unavailable portion of the plant (in DC MW peak)
- **NOMINAL POWER:** the nominal power of the entire plant (in kWp)
- **DATA DELAY:** the delay (in milliseconds) of data retrieval
- **ALARMS:** the number of detected PV Plant alarms
- **INVERTER STATUS:** the current operational status of PV Plant's inverter devices.

The platform uses a graphical informative pattern where four colored frames (green, yellow, orange and red) are displayed with the number of related inverters in their center. The color represents the status of the inverters: green for normal operation, yellow for operation under warning, orange for a significant fault, and red for a critical fault. All color patterns apart from green require the user's attention.

Finding a PV Plant in the List

in addition to scrolling through the list of sites, you may sort the list by any column's contents. Click on the any column header once for a forward sort, and again for a reverse sort. There is also a filter feature , approximately above the Data Delay column. This is active across the text fields and enables a search for any character string in these fields.

Figure 14. Using the Quick Find Button

Using the search filter will cause the list to present only the PV Plants sites that match the criteria, if any.

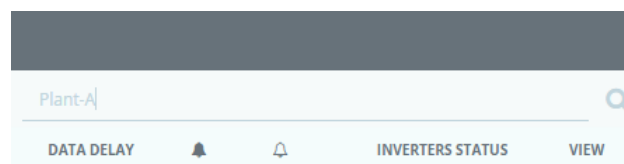


Figure 15. Successful search results

FLEET MANAGER												
STATUS	PLANT	COUNTRY	HUB	IRRADIANCE	POWER	PERFORMANCE	AVAILABILITY	UAL MWP	NOMINAL POWER	DATA DELAY	ALARMS	INVERTERS STATUS
70	PLANT-E	UK		LOADING -W/M ²	LOADING -KW	LOADING -%	LOADING -%		2,006.06	10.63	3	LOADING

Note that any search triggers an update of the information displayed. The update may take several seconds, as the derived values must be recalculated.

Figure 16. Site final presentation format of a PV Plant site



If you wish to select any site on the list for detailed information, simply move the mouse over the row of the desired plant. The row will be highlighted. If you click on the row, the application will move to the Plant Overview mode for that Plant. To return to the fleet list, select Fleet Manager under the Navigation Control icon.

The Cross Plant Export page

The second option in the Navigation Control panel is Cross Plant Export. This enables the fast collection and download of one or more data parameters of interest, from one or more supported PV Plant sites, in a comma separated value (CSV) or spreadsheet compatible .xlsx file. The ability to export a data set may be saved for future use, associated with the user account in use at the time, through the + CREATE button at the top left of the page. These templates can later be modified or deleted. To create a data export without saving the template, simply skip the left section of the screen and proceed with specifying the desired sites(s) and parameter(s).

A detailed instruction on the use of these commands follows.

Figure 17. The Cross Plant Export menu command

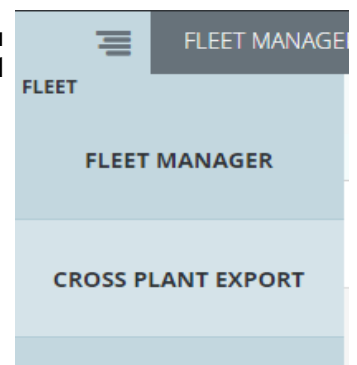
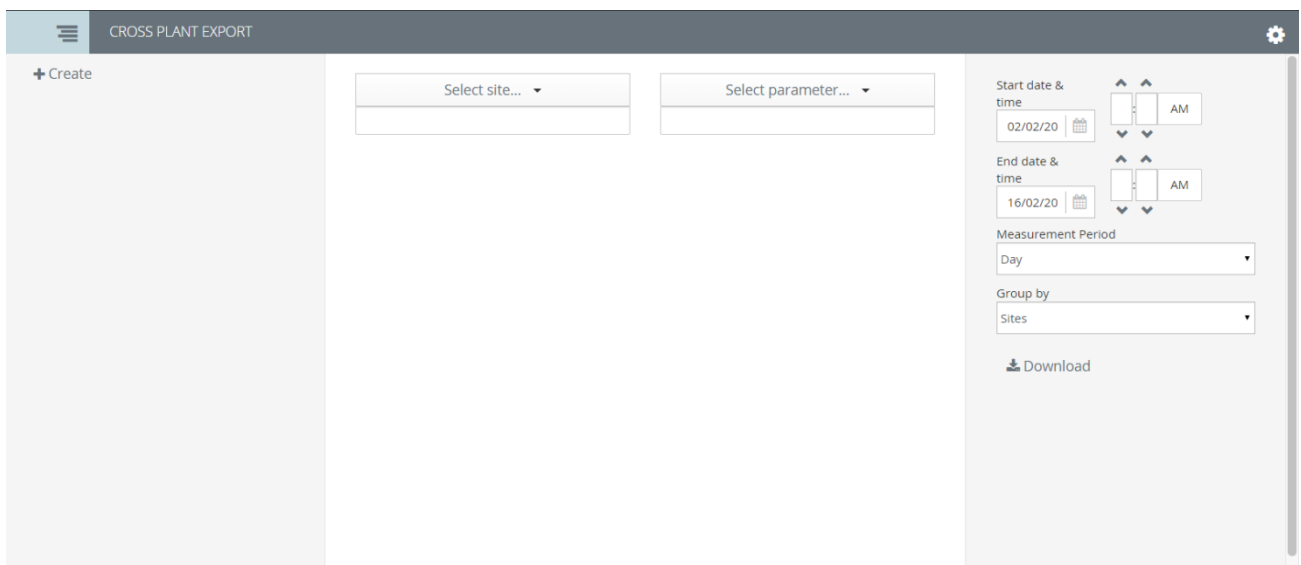


Figure 18. The Cross Plant Export page



Creating a new Cross Plant Export Template

Click on the “+ Create” button to create a new reusable export template. The next step is to enter a descriptive name for this new template.

Figure 19. Creating a new Template

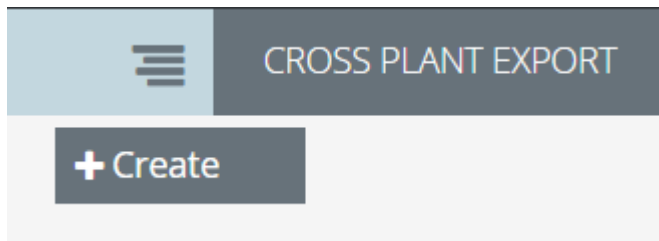
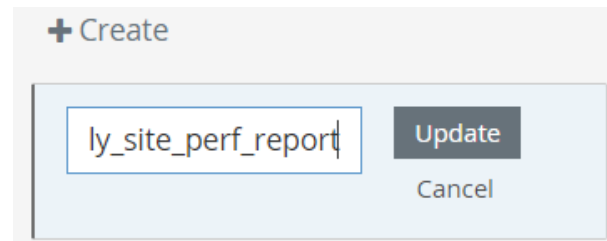


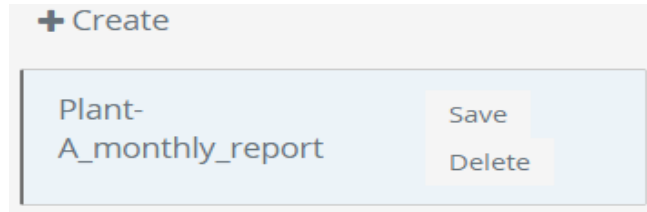
Figure 20. Naming the new Template



Upon clicking the “Update” button, the newly created template will be listed on the left side of the page within the pool of existing templates. The newly created template has not yet been stored into the system, so leaving this page will scrap the template. Next, define the PV Plant(s) and the Parameter(s) desired. Finally, press the “SAVE” (or “UPDATE”) button to retain your work for future use. Additional configuration parameters include Start Date & Time, Stop Date & Time, Measurement Period and Group By (sorting) parameters must be applied before and file can be exported. These are explained below.

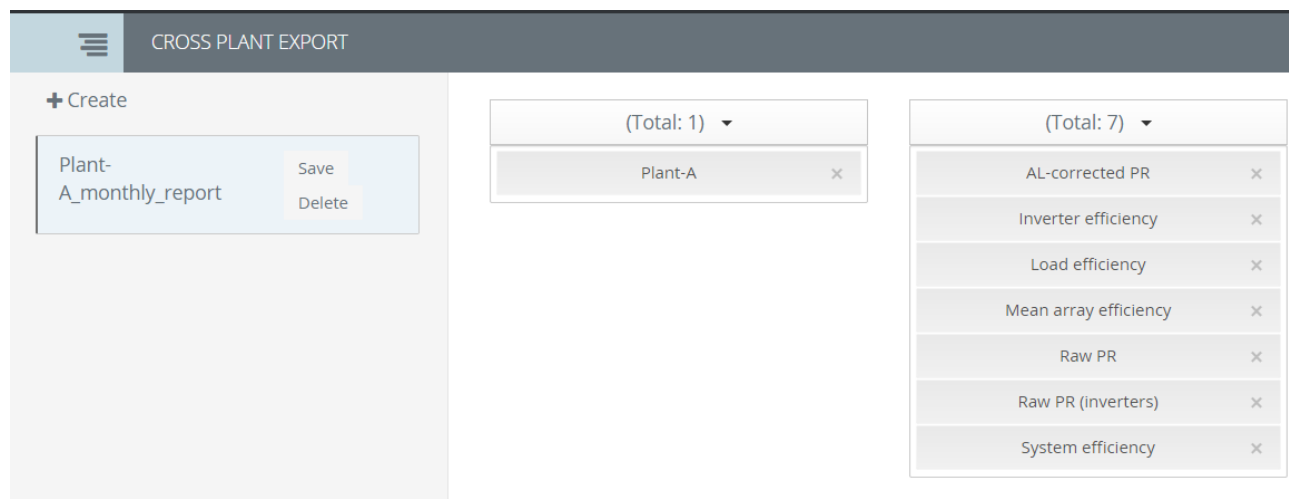
Note that the values are averaged over the site period selected. Site Period provides the finest resolution available through the download feature, typically 15-minute averages. Higher resolution data is available through the Reports feature.

Figure 21. The Template pool



Once the template is saved, the user may restore it selecting the template name selecting template from the existing pool. Any changes made during use may be saved with the UPDATE command.

Figure 22. Using an existing Template

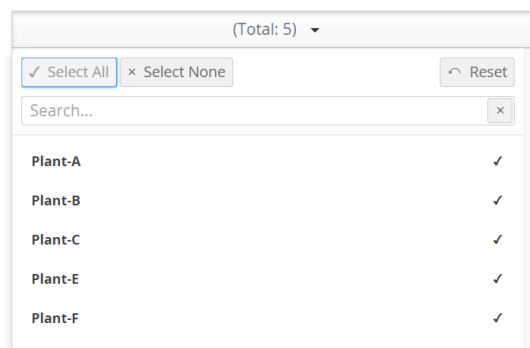


Configuring Cross Plant Export parameters

Unless your saved template has already done so, the first step is to specify the Plant(s) and Parameter(s) desired for the report.

In both the Plants and Parameters columns, you may select or deselect items by clicking on them. First open the column by selecting the header. The relevant options are then displayed. “Select All”, “Select None”, and “Reset” options exist, as well as a search filter. To use the search filter, enter the desired term. It is best to use a short string and select from many options, rather than to try to completely define the exact parameter desired. Click on the desired choices in the filtered list. To continue selecting from a different filtered list or the master list, simply change the search string or remove it with the X icon to the right of the search term box.

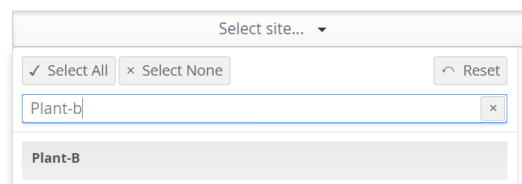
Figure 23. Selecting corresponding PV Plants by using the “Select All” button



The screenshot shows a selection window titled "(Total: 5)". At the top, there are three buttons: "✓ Select All" (highlighted with a blue border), "× Select None", and "↶ Reset". Below these is a search bar labeled "Search..." with a clear "X" icon. The main area is a list of plants, each with a checkmark to its right:

Plant	Selected
Plant-A	✓
Plant-B	✓
Plant-C	✓
Plant-E	✓
Plant-F	✓

Figure 24. Selecting corresponding PV Plant using the search control



The screenshot shows the same selection window, but the search bar now contains the text "Plant-b". The "Select site..." dropdown at the top is also visible. The list of plants is filtered to show only "Plant-B", which has a checkmark to its right.

Plant	Selected
Plant-B	✓



Figure 25. Selecting corresponding Parameters by using the “Select All” button

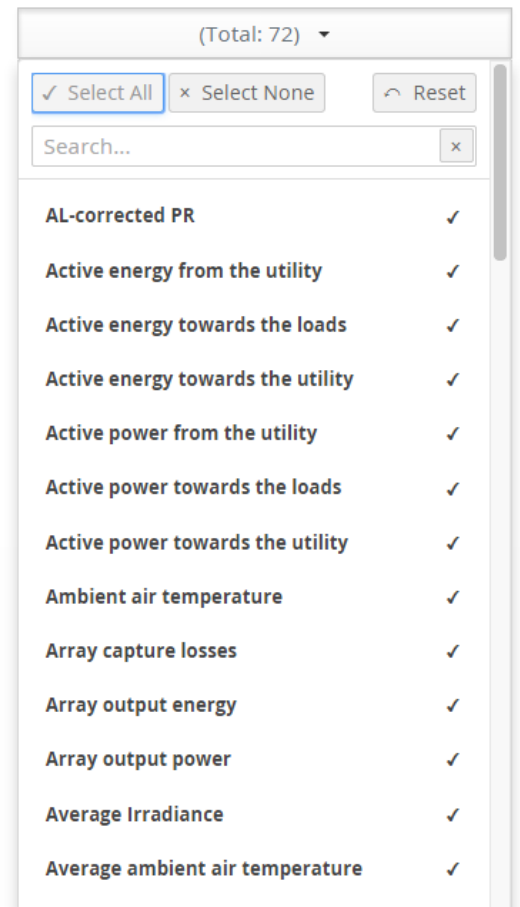
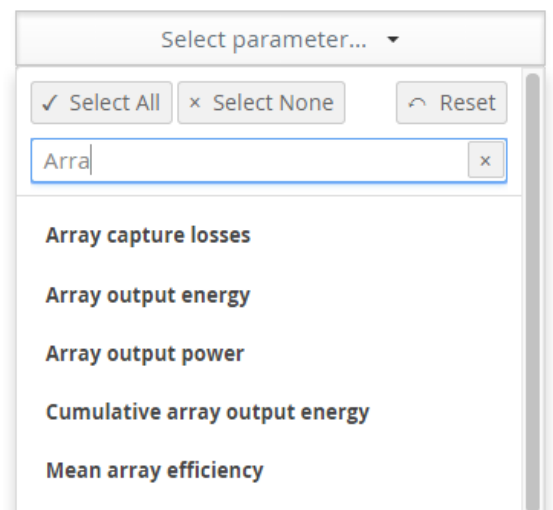


Figure 26. Selecting corresponding Parameters using the search control



Next, specify the remaining data collection parameters for your download as follows:

- **Start and Stop Date:** Define the date range for the report.

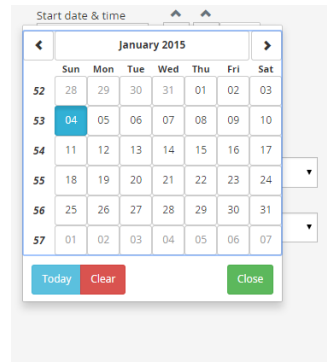


Figure 27. Specifying “Date” parameter

- **Start and Stop Time:** Enter hours and minutes. The AM/PM section is a toggle.

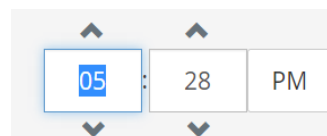


Figure 28. Specifying “Time” parameter

- **Measurement Period:** Refers to the Period over which data is averaged or totaled, as appropriate the parameter. Options are Site Period, Day, Month and Year. Site Period is the “bin size” for measurements from the site, typically 15 minutes.

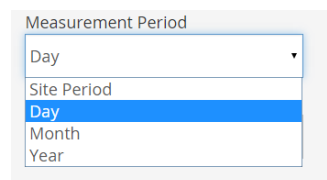


Figure 29. Specifying “Measurement Period”

- **Group By:** Arrange the extracted results by Parameter type or by Site.

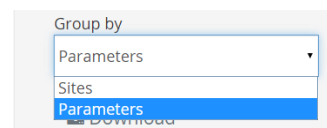


Figure 30. Specifying “Group By” parameter

Finally, choose between CSV and .xlsx formats for the file. Click download to bring the file to your computer. Note that the location of the download file is set by your computer’s settings, normally either the Downloads folder or your Desktop.

+

Create

Plant-A_monthly_report

Save

Delete

(Total: 1) ▾

Plant-A ✕

(Total: 7) ▾

AL-corrected PR ✕

Inverter efficiency ✕

Load efficiency ✕

Mean array efficiency ✕

Raw PR ✕

Raw PR (inverters) ✕

System efficiency ✕

Start date & time

17/02/20 

End date & time

01/09/20 

Measurement Period

Month ▾

Group by

Parameters ▾

Download 

The Report Subscriptions page

The third option on the Navigation Panel leads to the report services that the platform offers. Through this option, a variety of one-time and periodic reports are available for delivery to the email address registered in the user profile.

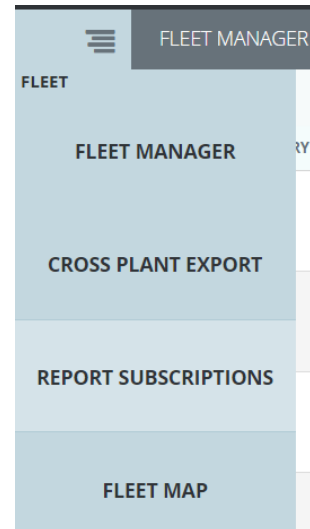


Figure 32. Using the “Report Descriptions” command button

The “Report Subscriptions” page presents a chronologically sorted list of the report subscriptions associated with the login. The list is sortable by any of the column headers, namely Report Name, Plant, Report Type, Report Format, Report Interval, and Status.

Figure 33. Using the Report Subscriptions page for first time



To create a new report subscription, use the “+Create” button placed at the top-right of the header panel.

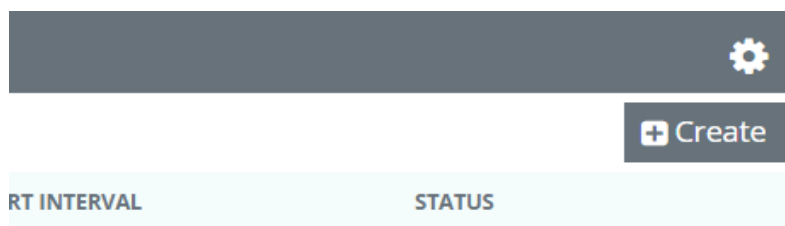
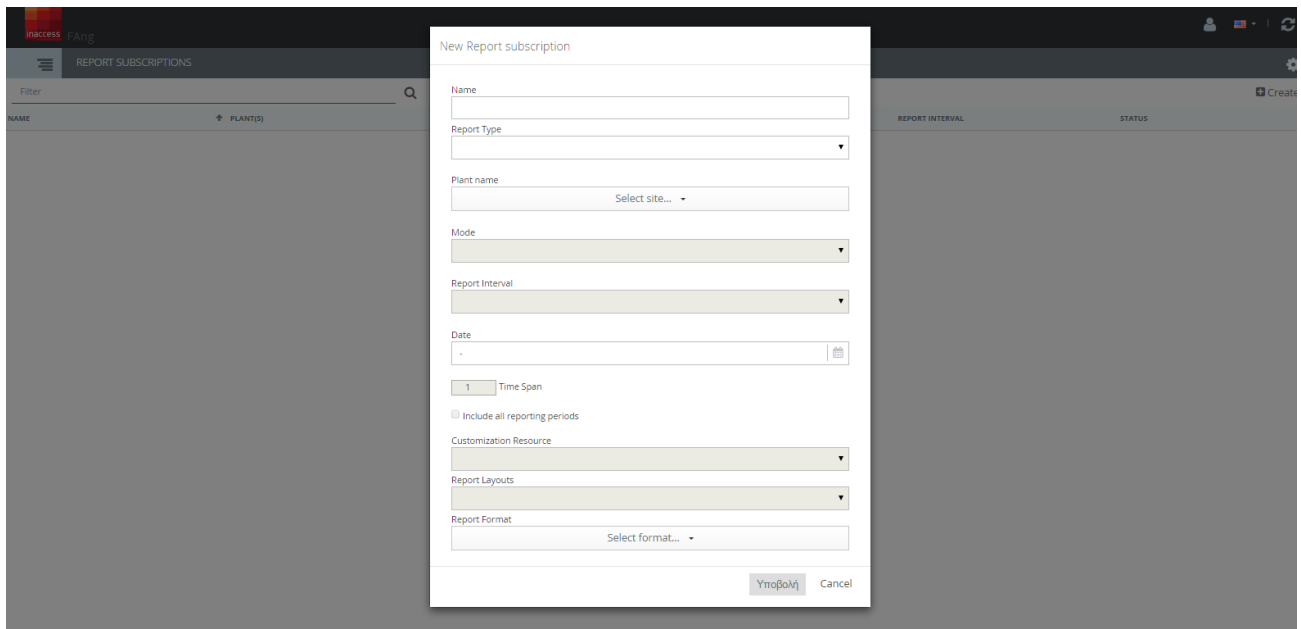


Figure 34. Using the “+Create” button

The “+Create” button activates the “New Report Subscription” dialog frame, which is used to fully specify the report and the terms of its subscription.

Figure 35. The “New Report Subscription” dialog



If you intend to use a particular report for regular updates, it is helpful to give the report a descriptive name. Include the plant name and perhaps the purpose of the report or its key contents. If, however, you are making a one-time download, you might include a reference number to uniquely identify the report.

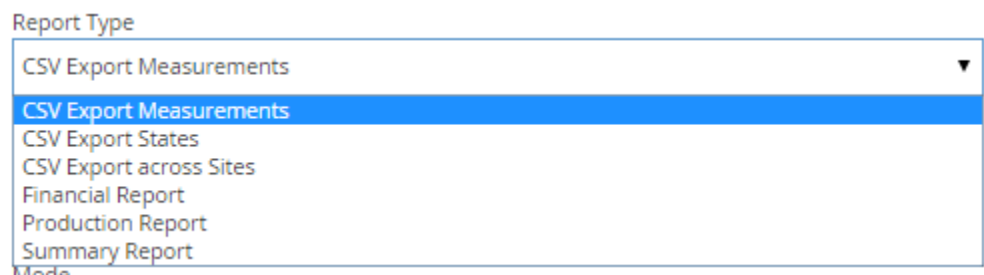
Figure 36. The “Name” field



The “Report Type” drop-down menu offer six choices, explained below.

Figure 37.

The “Report Type” field





➤ **CSV Export Measurement:** This report type provides user-defined measurements. The “Customization Resource” later in the window makes use of pre-defined sets to form the basis of the report. The User first has the ability to request for measurements from various levels in the hierarchy of the site. (The top or Site level, PCC, TCP, ArrayGroup and Array levels are options.) Measurement type list can also be enhanced with additional customized measurement templates that the User can apply to the system with the assistance of “Scratch Pad” functional service that will be explained in the “Preview Tab” section of the “Browser” page clause.

Customization Resource

All parameters of segments Site (preset)
 All parameters of segments PCC (preset)
 All parameters of segments TCP (preset)
 All parameters of segments ArrayGroup (preset)
 All parameters of segments Array (preset)

Figure 38. The corresponding “Customization Resource” control for “CSV Export Measurement”

➤ **CSV Export States:** This report type provides operational status information, the resource type of which will also be specified with the assistance of “Customization Resource” control. The User is able to distinguish between several state resources categories covering Site. PCC. TCP, ArrayGroup and Array segment penetration levels.

Customization Resource

All states of segments Site (preset)
 All states of segments PCC (preset)
 All states of segments TCP (preset)
 All states of segments ArrayGroup (preset)
 All states of segments Array (preset)

Figure 39. The corresponding “Customization Resource” control for “CSV Export States”

➤ **CSV Export across Sites:** Such reports offer the ability to correlate measurements extracted by two or more PV Plants. The User can additionally specify the “Customization Resource.” There is the choice of a standard monthly report and an in-detailed report covering every plant level parameter.

Customization Resource

Plant-A_monthly_report
 All Plant Level Parameters (preset)

Figure 40. The corresponding “Customization Resource” control for “CSV ExportAcross Sites”



- **Financial Report:** A pre-defined report type that captures the financial performance of the plant, offering a set of related information and graphs. The “Customization Resource” is not used for this type of configured report.

Customization Resource

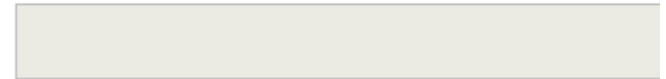


Figure 41. The corresponding “Customization Resource” control for “Financial Report”

- **Production Report:** A pre-defined report type that delivers performance metrics of the plant, offering a set of related information and graphs. The “Customization Resource” is not used for this type of configured report.

Customization Resource



Figure 42. The corresponding “Customization Resource” control

- **Summary Report:** A more extensive pre-defined report type that covers financial results, energy production totals, and the performance ratio of the plant. The “Customization Resource” is not used for this type of configured report.

Customization Resource



Figure 43. The corresponding “Customization Resource” control

The next step is to select the PV Plant site used for the report. All report types deal with a single PV Plant, however the “CSV Export Across Sites” and “Summary Report” also enable reporting from multiple Plants, including from the entire fleet of plants.

Figure 44. The “Select site” parameter

Plant name

Select site... ▾

Reset

Search... x

Plant-A

Plant-B

Plant-C

Plant-E

Plant-F

The “Mode” option offers a one-time “On-demand” or a recurring “Scheduled” option. On-demand report “subscriptions” are deleted once the report is produced and sent. Depending on the complexity of the report and the date range selected, on-demand does not mean instantaneous.

Mode

On-demand

Scheduled

Figure 45. The report “Mode” parameter

Report Interval

Daily

Weekly

Monthly

Yearly

The “Report Interval” parameter offers “Daily”, “Weekly”, “Monthly” and “Yearly” options.

Figure 46. The “Report Interval” parameter

Date

01/06/2014

Excepting “Scheduled” report modes, the date for the report data is specified next. In the scheduled mode, the report template the term is defined by the subscription period.

Figure 47. The “Date” parameter

At this point it is possible to specify the insertion of additional periods into the template. For example, a subscription to a daily report including 7 time spans would include the current day and the six previous.

Also shown here is the very powerful “include all reporting periods” option. This feature provides the maximum level of detail, with the full set of data points on file for the requested parameters. Typically,

this would be one-minute data. Start with very short reporting periods to avoid excessively large data sets.

Figure 48. Additional time definition parameters

☒ Fixed Start Time

Time Span

☒ Include all reporting periods

Finally, you have the choice of comma separated value (csv), excel (xlsx), open document template (odt) and pdf file formats within “Report Format” control. (Note that the CSV data is actually separated by semicolons, not commas. Commas are used as decimal points in some localizations; semicolons eliminate any chance of conflict. Enabling semicolons as the separator is an option in the spreadsheet to which the data is imported.)

Figure 49. The “Final Format” parameter

Report Format

Select format... ▼

Search...

CSV

XLSX

Request your report by selecting “Submit Query.” The report will be added to your report subscriptions list. You may cancel it by clicking on the trash can icon at right.

Figure 50.

The Report Subscriptions List

REPORT SUBSCRIPTIONS					
NAME	PLANT(S)	REPORT TYPE	REPORT FORMAT	REPORT INTERVAL	STATUS
Financial_Report	Aprilia	financial	Html,Odt,Pdf	20141201	OK
Production	Aprilia	production	Html,Odt,Pdf	20150101	OK
Summary	Aprilia	summary	Html,Odt,Pdf	20150101	OK
Export Measurements	Aprilia	csvexportmeasurements	CSV,XLSX	20150101	OK
Export States	Aprilia	csvexportstates	CSV,XLSX	20150101	OK

The Fleet Map page

The Fleet Map offers a visual representation of every PV Plant visible to the user's account. Plant representations are based on its geolocation parameters, and the map can be manipulated with the Google Map controls.

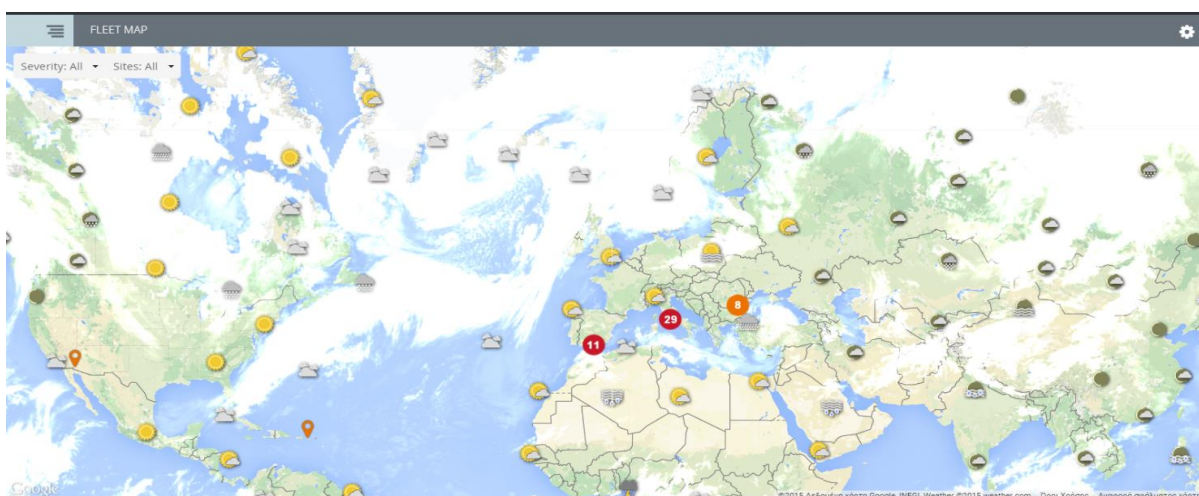
Plants are displayed with map pins. Multiple Plant sites which graphically overlapping each are displayed with a white number on in their pin icon indicating the number of PV Plant sites located on the same region. The color of this pin icon depends on the highest alarm status (highest severity level) that may occur in a PV Plant site of the whole group.

Figure 52. PV Plant Grouping Icon samples



Figure 51. Using the “Fleet Map” command button

Figure 53. The “Fleet Map” page



The Fleet Map also offers real-time meteorological data to give a clear overview about the weather conditions in a specific geographical region.

You can also further inspect a PV Plant site of interest by simply clicking the mouse on the associated pin-icon. This action delivers a popup tooltip that holds the most essential informative parameters of the selected PV Plant site. This information covers the Name of the site, the existence of an active alarm, the site nominal and produced power, the site availability and its elementary financial parameters.

Figure 54. PV plant site tooltip activation

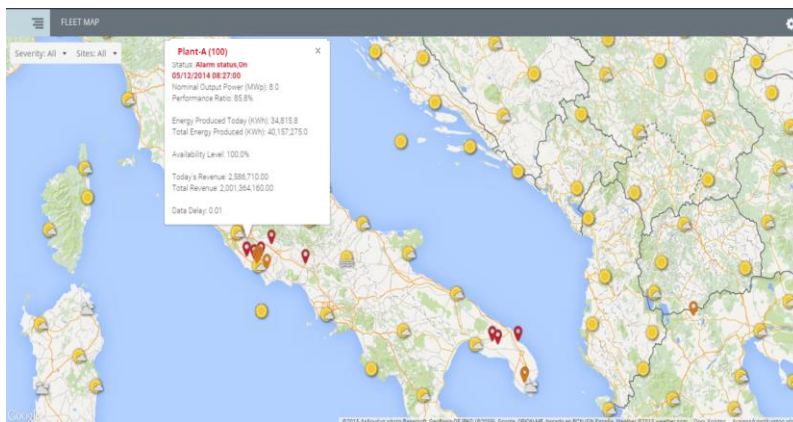
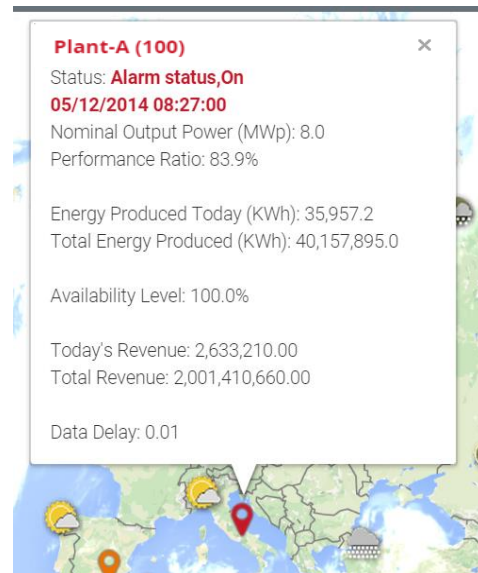


Figure 55. A typical PV Plant site informative tooltip



This page embeds two filtering controls placed on the topmost-left side of the control bar. The first allows filtering by operational status: normal, warning, fault and critical fault. The second allows for the selection of specific plants by name.

In cases where more than one operational status have been checked system displays the sum of PV Plant sites that matches these statuses coloring the illustrated pin icon with the color of the most critical status.

Figure 56. Embedded Operational Status Filtering Options

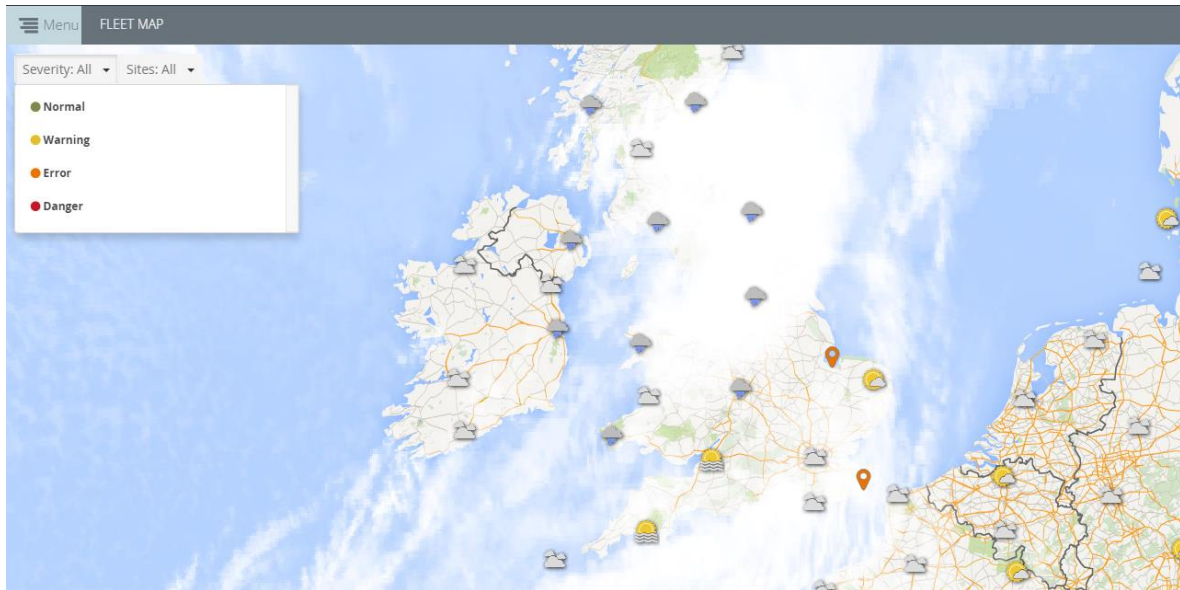


Figure 57. Embedded PV Plant Search and Selection options

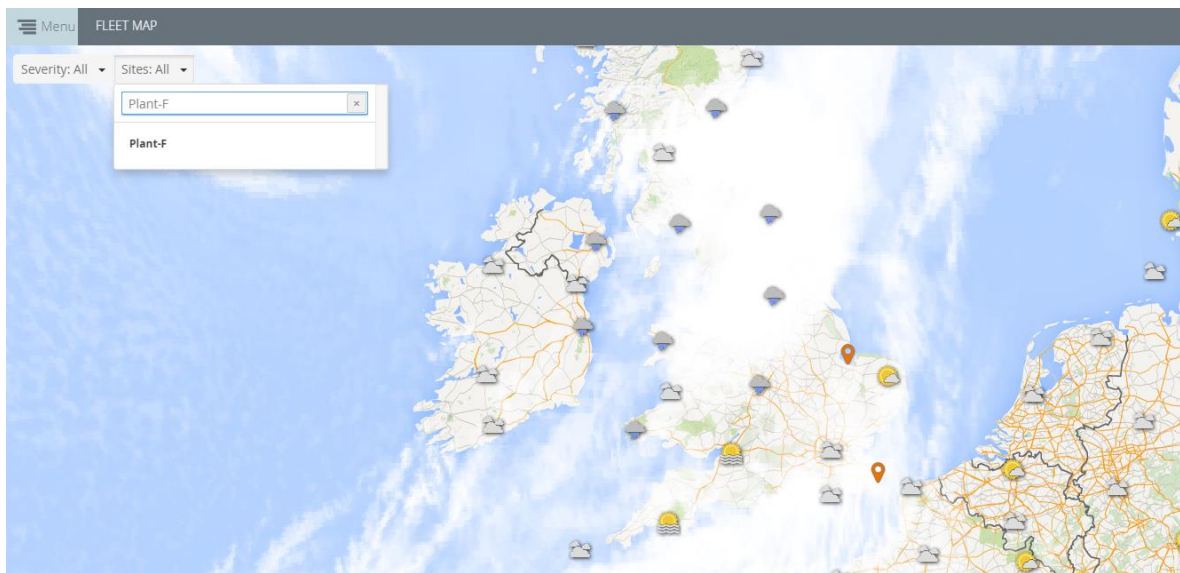
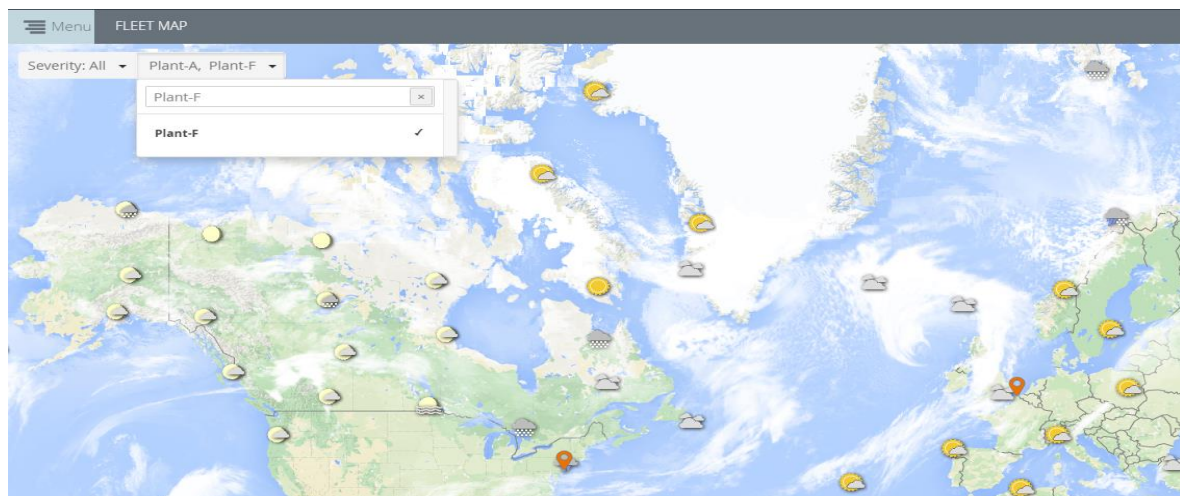


Figure 58. Automatic display of a selected PV Plant site



The Sites Tree Map page

“System Tree Map” is a newly introduced feature which offers a comprehensive conceptual overview regarding the operational status for the whole fleet of PV Plants.

This functional feature is based on CIM introduced Hierarchical Status Propagation (HSP) philosophy which is a hierarchy model that aims to define how state events travel from segment to segment and from asset to segment in a hierarchical manner (tree structure) with specific constraints such as the root of the tree is always the site itself, sub segments like PCC, TCP, LCP, Inverters and any other are children nodes of this root and every coherent asset is a leaf of these tree nodes.

The site trees are not presented in the classical form of a SCADA-like tree, rather in an abstract mode designed to emphasize the significance and severity of status conditions. Green, yellow, orange, and red are used consistently within the inaccess platform. The size of the rectangles in the presentation corresponds to the production capacity of the asset.

By clicking on and thus expanding any rectangle, the next deeper level of equipment is exposed. The position in the hierarchy is given below the Navigation Control icon. It is thereby possible to graphically follow the source of fault conditions to their root.

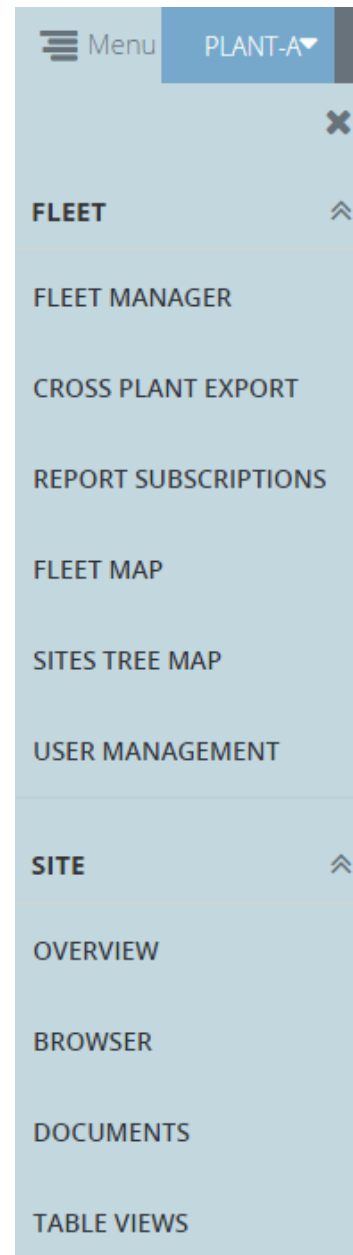


Figure 59.
Using the
“Sites Tree
Map”
command
button

Figure 60. The Header Sites Tree navigation control



Figure 61. The main “Sites Tree Map” page



In order to fully comprehend the represented decomposition rectangle mosaic you should be familiar with the structural form of a PV Plant site. In general each site is consisted of individual functional segments (logical and virtual functional groups) like:

- **PCC:** the Point of Common Coupling, which refers to the physical interconnection point between plant's local power-line with the external grid. This is the point where all requirements mandated on the PV plant by the EPS operator. Only one PCC can exist per PV plant.
- **IT:** the local controlling device(s) and assistant computing infrastructure for the proper intercommunication of central management system (back-end) with local devices and the propagation of monitored measurements and control events.
- **Equipment:** Any additional equipment (outside of the scope of the aforementioned categories) that may assist aggregation of additional monitoring information implied by existing sensor devices.
- **Weather Stations:** Sensor devices that assist the real time monitoring of environmental conditions.

The site segment decomposition scheme may vary between different sites according to planning model and commissioning alternations or restrictions.

Figure 62. Selecting a PV Plant decomposition segment

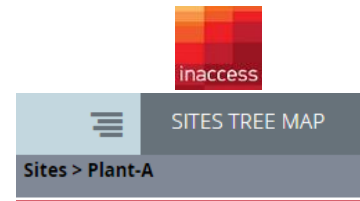
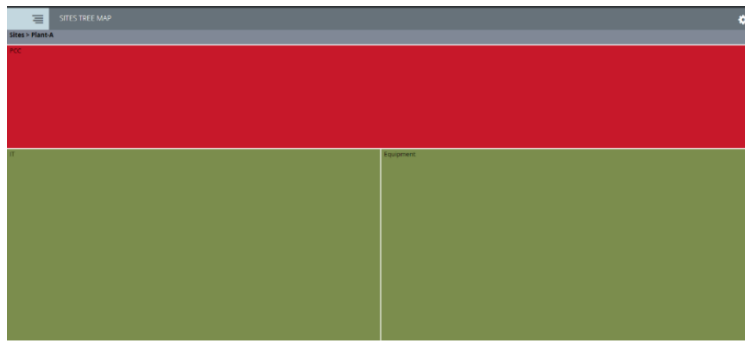


Figure 63. The corresponding header navigation control

Upon selecting a PV Plant site of interest the header link control is reformed in order to display the referred site name in the form of a parent to descendants string (Parent>Descendant>Descendant of Descendant.....).

The header string is through an active navigation control and you can, by clicking on a string part, move on the corresponding penetration level where in parallel the page will be immediately refreshed in order to present the decomposition entities of the selected segment and the header link control will also reformed to display the new navigation sequence, respectively.

Figure 64. Selecting a PV Plant decomposition segment indicating an error



Figure 65. The corresponding header navigation control

You have the ability to repeatedly follow the same inspection pathway passing through information levels of interest until you find the reason and the corresponding to the indicated erroneous event or alarm asset.



Figure 66. Selecting an Array Group indicating an error

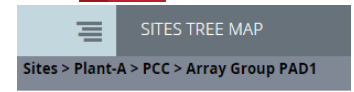
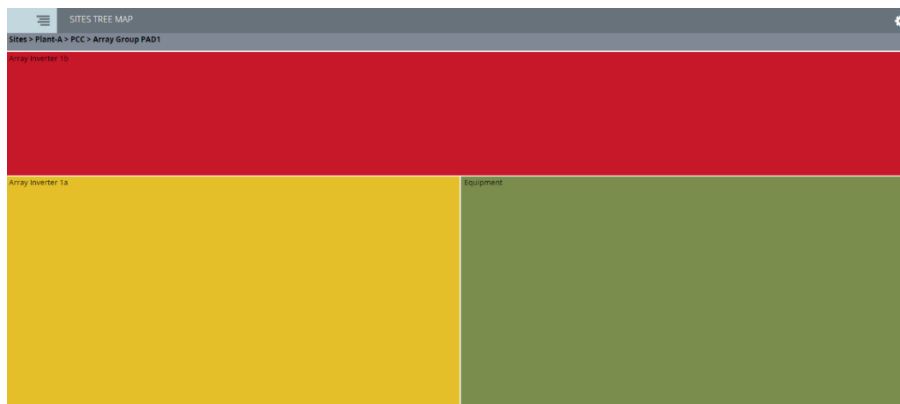


Figure 67. The corresponding header navigation control

Figure 68. Selecting an Array Inverter indicating an error

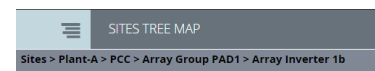
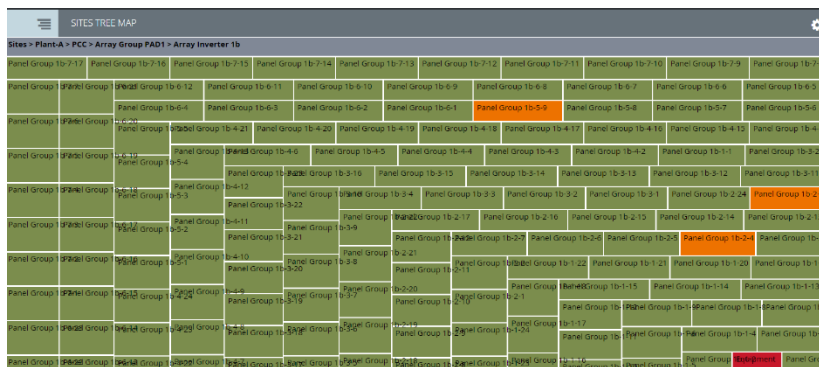


Figure 69. The corresponding header navigation control

Figure 70. Detecting the Equipment that triggers the error

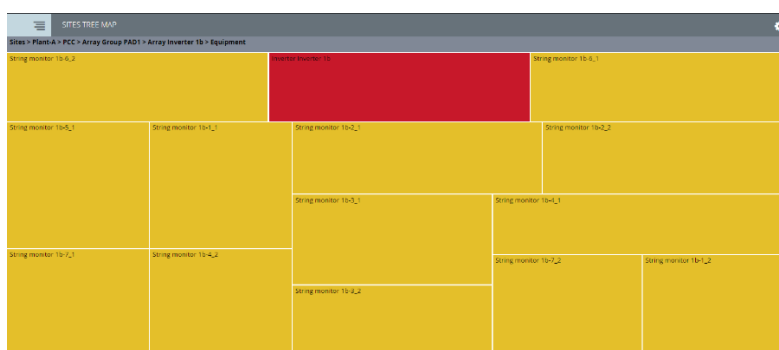


Figure 71. The corresponding header navigation control

The User Management page

The Inaccess web portal application is intended for many types of users with an interest in PV Plants. These may include site maintenance contractors, owner, the operational team, security agents or investors. One of the users must be an Administrator. Only the Administrator has access to the User Management page.

Administrators have the ability to **Add** new User profiles, to **Modify** existing User Profiles and to **Assign** (or associate) User profiles with PV Plants.

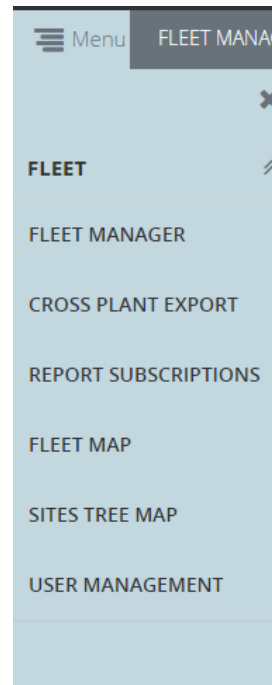


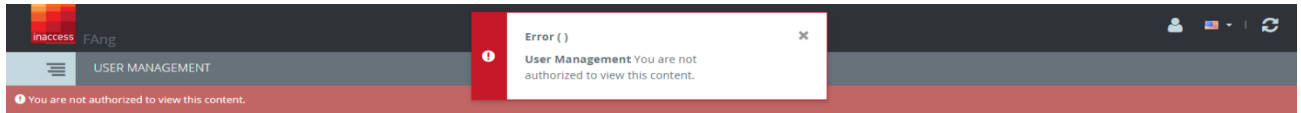
Figure 72. Using the “User Management” command button1

Figure 73. User Management account overview

USER MANAGEMENT								+	+
USER NAME	↑	SURNAME	AGENT TYPE	MOBILE PHONE NUMBER	E-MAIL	PHONE NUMBER	ACTIVE		
admin			System Administrator		-	-	Yes		
akostop	Alexandros	Kostopoulos	System Operator		akostop@inaccessnetworks.com		Yes		
ALM-AH1			IFMS S/W module		-	-	Yes		
ALM-D1			IFMS S/W module		-	-	Yes		
ALM-D2			IFMS S/W module		-	-	Yes		
ALM-M1			IFMS S/W module		-	-	Yes		
ALM-TR1			IFMS S/W module		-	-	Yes		
ALM-TR2			IFMS S/W module		-	-	Yes		
ALM-TR3			IFMS S/W module		-	-	Yes		
ALM-Y1			IFMS S/W module		-	-	Yes		
alykouras			System Administrator		-	-	Yes		
amanous	Angelos	Manousarides	System Administrator	306946177490	amanous@inaccessnetworks.com	302106802352	Yes		
arisan	Aris	Anagnostou	System Operator		arisan@inaccess.com	-	Yes		
backend			IFMS S/W module		-	-	Yes		
bererekon			System Operator		-	-	Yes		
cfo			System Operator		cfo@inaccess.com	-	Yes		
cgeorg	Christos	Georgopoulos	System Operator	00306944634286	cgeorg@inaccess.com	00302106802300	Yes		
cgeorg_srp			System Operator		-	-	Yes		
ciolas	Christos	Lolas	System Administrator	306976440194	ciolas@inaccessnetworks.com		Yes		
cmeletis	Christos	Meletis	System Operator	306934088417	cmeletis@inaccessnetworks.com		Yes		
cpats	Christos	Patsiouras	System Administrator		cpats@inaccess.com	-	Yes		
cperaki	Christina	Peraki	System Operator		cperaki@inaccessnetworks.com		Yes		

A user without administrative privileges will receive the following error message.

Figure 74. The User Management Access Restrictions



Creating a new User Profile

System administrators have the ability to create new User profiles by selecting “+Add User”. This will open a dialog frame where the new account information can be entered.

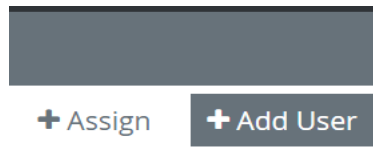


Figure 75. Creating a new User profile

Each User profile record contains the following parameters:

- Name: the permanent user login name for this account
- Access Rights: choice of six levels of permission
- Password: the user password, required during login process
- Confirm Password: verifying the password
- First Name: first name of the new user
- Surname: last name of the new user
- Mobile: mobile phone number of the user, used for the integration of SMS Notification services
- Email: the email account of the new User, used for the integration of Email and Report Subscription and Notification services
- Phone Number: second phone number for the user
- User Role: the functional role of new User that also specifies the PV Plant information accessibility level
- Active: the option to inactivate a user account
- Sites: the PV Plants made visible to the account.

The Access Rights parameter is a key element of the user account. This determines the ability to access specific groups of supported services such as the planning summary, plant management, plant browsing, subscription to SMS/Email notifications, report creation, and user management services.

Figure 76. Specifying new User profile Parameters

Figure 77. Specifying new User Access Rights

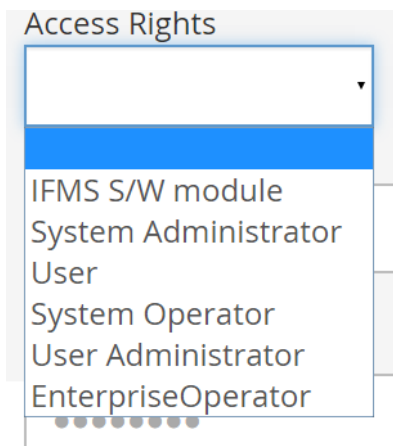


Figure 78. Specifying the User Role

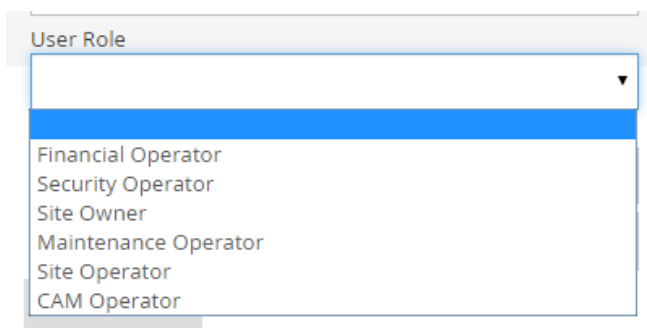
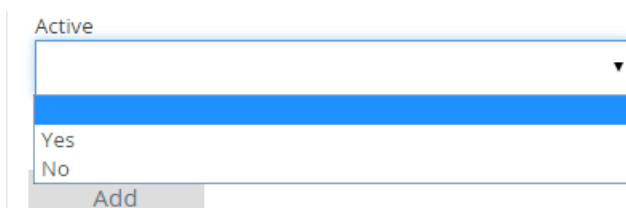
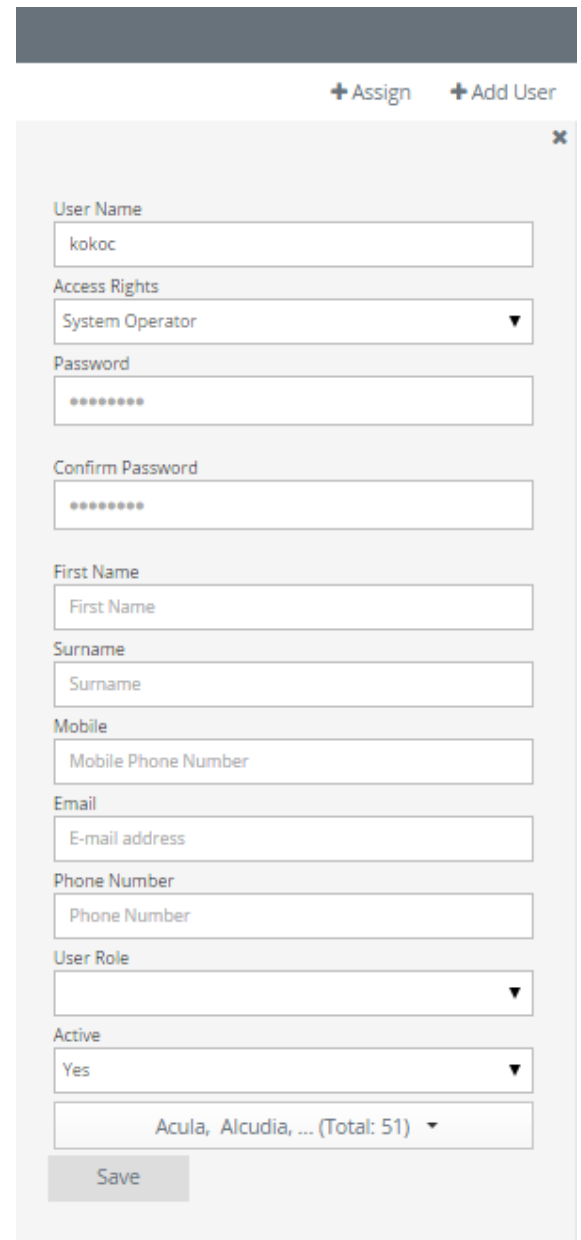


Figure 79. Specifying the User Activity Status

The user profile must be recorded with the “Save” button. The newly defined user profile will then be attached to the records of existing profiles in the main page list. The X button cancels the account creation.

Modifying Existing User Profile

Whenever a user with administrative privileges enters the User Management page a list of existing profiles is automatically displayed. To modify a profile, click on the profile of interest. The same dialog frame used in account creation appears.

Figure 80. Modifying a User Account

+ Assign + Add User

	ACTIVE	X
No	User Name gstefan	
No	Access Rights System Administrator ▼	
No	Password ● ● ● ● ● ● ● ●	
No	Confirm Password ● ● ● ● ● ● ● ●	
No	First Name First Name	
No	Surname Surname	
No	Mobile Mobile Phone Number	
Yes	Email gstefan@inaccess.com	
Yes	Phone Number Phone Number	
Yes	User Role ▼	
Yes	Active Yes ▼	
Yes	Acula, Alcudia, ... (Total: 51) ▾	
Yes	Save	

Assigning a User Profile to a PV Plant Site

The System administrator has also the ability to assign to multiple users access to any PV Plant in one operation, without needing to perform this action repetitively for each user. use the “+ Assign” button and select your plant, then the role level, and then the users to be associated.

Figure 81. Activating the PV Plant site – User Profile Association Procedure

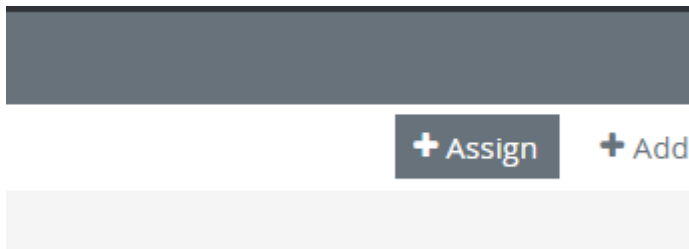
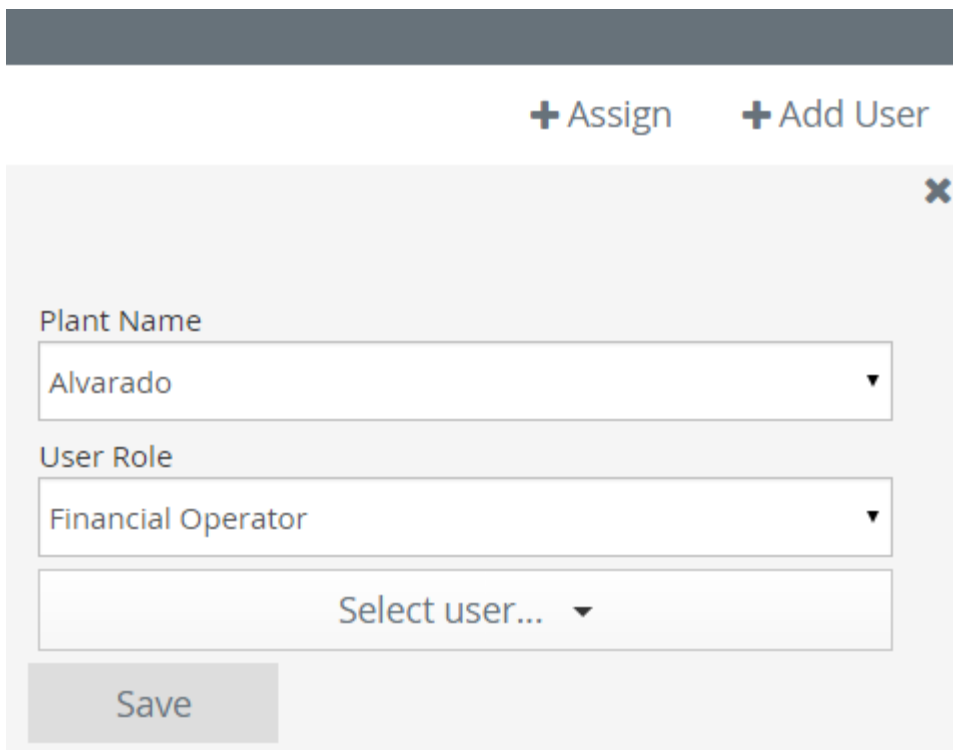


Figure 82. The PV Plant site – User Profile Association Dialog Frame



Site Level: The Overview page

After selecting a site of interest from the Fleet Manager, Fleet Map, or elsewhere, the options of the Site Mode are enabled. The first page to open is the “Overview” navigation page. This is an explanatory overview of the selected site. It consists of four individual horizontal layers. The default configuration includes all the key functional parameters of the site in numerical, graphical and lists formats.

The information displayed updates automatically. An update may also be forced with the chasing arrows icon to the right of the language flag.

It is possible to return to this page from elsewhere in the application by selecting “Overview” under the “Site” portion of the Navigation Control menu.

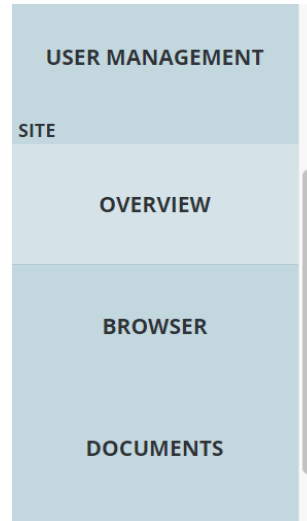
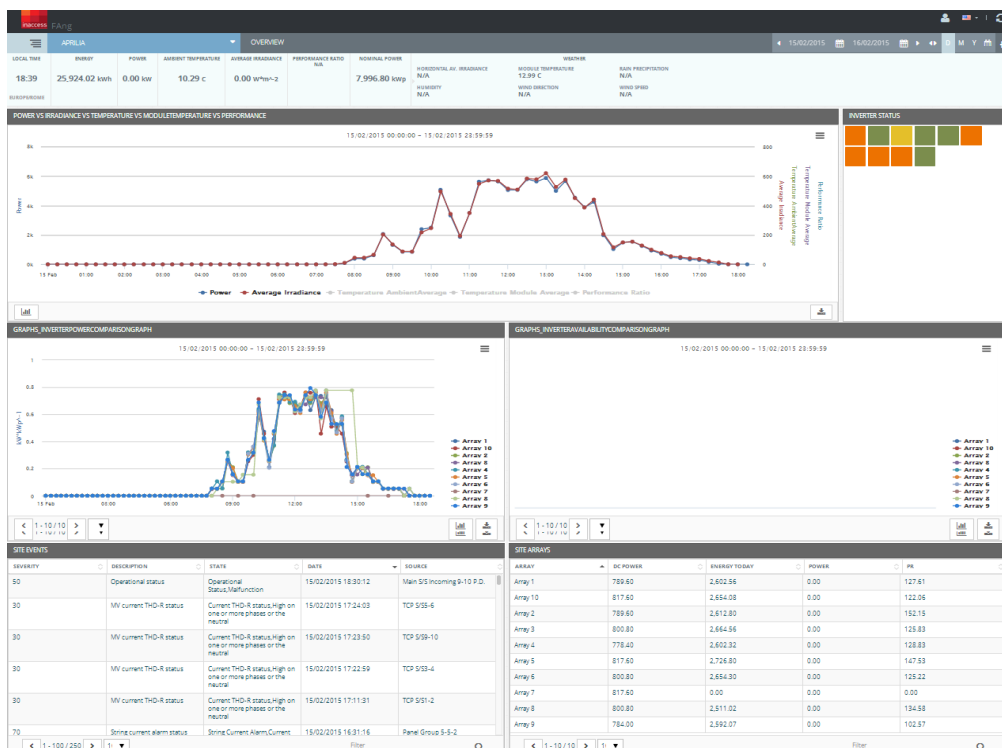


Figure 83. Using the “Overview” command button

Figure 84. The “Overview” page



Dynamic Header Site Controls Layer

At the upper right, under the user icon and language selection flag, are the time controls and a settings icon. These affect most of the displayed information on the Overview page, aside from the current parameters described below.

The settings icon (gear) at the end of the bar allows for reconfiguration of the display, and is addressed in the section titled Optional Graphs.

Upon first viewing the page, the D, M, and Y and calendar icons are set to the current day, current month (from the first until today), current year (from January 1 until today), and (once again, in the calendar icon) the current day, respectively. You may easily change from the current day by using the calendar icon at the right to select a new day. Also, the right and left arrows will step the time forwards or back in increments of the unit being shown. The outward facing pair of arrows increases the range of views by one increment at each end (up until the present moment, of course). Furthermore, you may select specific start and end dates with the calendar icons next the right and left arrows.

It's best just to explore the function of the various commands and become comfortable with their uses. Recall that you have the option of dd/mm/YYYY or mm/dd/YYYY date formats in your user profile. Just click on the user icon next to the language flag to find it.

Note that the graphs also feature the ability to zoom in on an area of interest by clicking a corner and dragging a window over a portion of the data. Mousing over the new window reveals a pop-up command to "Reset Zoom" and leave the subset area.

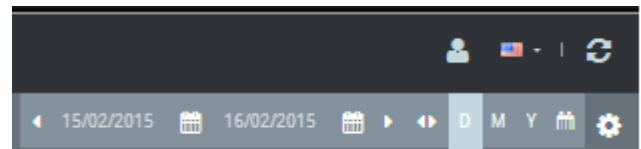


Figure 85. The Dynamic Header Site Controls

Numerical Site Information Parameter Layer

The first horizontal layer in the display provides a collection of current key environmental and efficiency values. As these are the most up-to-date reported values, the time controls have no effect on their display.

Figure 86. Numerical Site information Parameters

LOCAL TIME	ENERGY	POWER	AMBIENT TEMPERATURE	AVERAGE IRRADIANCE	PERFORMANCE RATIO N/A	NOMINAL POWER	WEATHER		
18:39	25,924.02 kwh	0.00 kW	10.29 c	0.00 W/m ²		7,996.80 kWp	HORIZONTAL AV. IRRADIANCE N/A	MODULE TEMPERATURE 12.99 C	RAIN PRECIPITATION N/A
EUROPERDOME							HUMIDITY N/A	WIND DIRECTION N/A	WIND SPEED N/A

- **LOCAL TIME:** The local time at the PV Plant
- **STATUS:** The most significant event alert presently at the Plant and the affected equipment.
- **ENERGY TODAY:** The cumulative produced energy (in KWh) from midnight today until now
- **POWER:** The present power output of the plant (in kW)
- **AMBIENT TEMPERATURE:** The ambient air temperature at the PV Plant
- **POA:** The plane of array irradiance value, averaged from all dedicated sensors if more than one exists.

- **PERFORMANCE RATIO:** The overall performance ratio of PV Plant (%)
- **NOMINAL POWER:** The nominal size of the plant (DC kWp)
- **WEATHER:** Six environmental factors, displayed if available from local sensors. HORIZONTAL IRRADIANCE (averaged if from multiple sensors), HUMIDITY, MODULE TEMPERATURE, WIND DIRECTION, PRECIPITATION and WIND SPEED.

The Overview Dashboard

The remainder of the Overview page is composed of a set of **tabs** containing informational graphs or tables. Users have the ability to resize, reorder, or remove tabs as desired. The default Overview page employs a group of **primary** tabs which are the most common metrics for plant performance and health. There is also a pool of **optional** tabs available, providing additional monitoring functionality. You have the ability to attach/detach any type of tab using the Configuration command. (Consult the section titled “Configuring Overview Displaying Options” for more details.) It is always possible to return to the original defaults.

Standard Graphs

The Main Performance Site Graph Tab

The first graphical representation in the default configuration displays PV Plant site performance. Power production is graphed with Average Irradiance. Further options include Ambient Temperature and Module Temperature parameters, and Performance Ratio. Click on each variable in list below the graph to include or exclude it from the graph. The time controls at the top of the page affect this graph. and the zoom-to-area feature is available here.

Figure 87. Power, Irradiance, Ambient and Module Temperature: Line style control

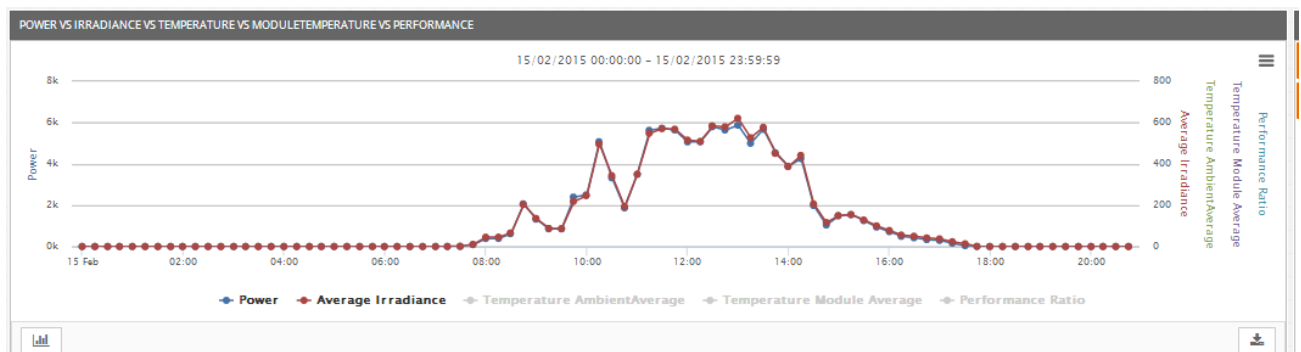


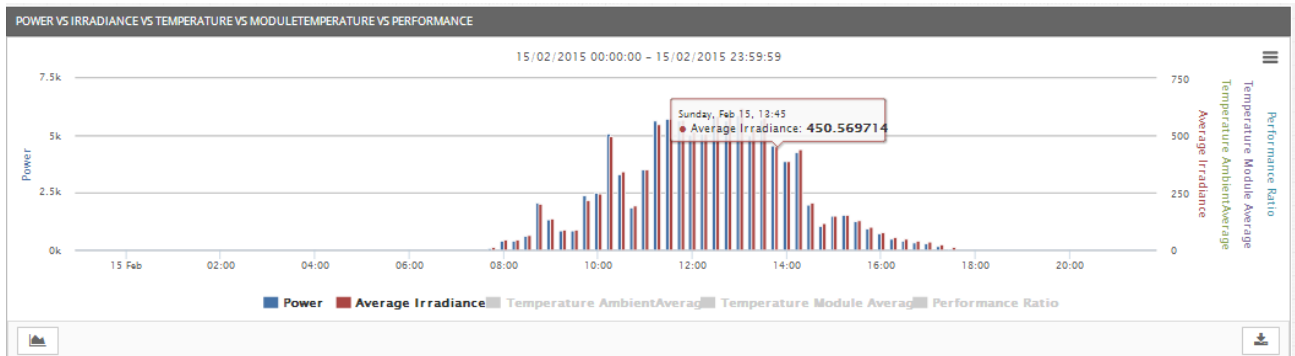
Figure 88 Performance Graph Available Parameters

☒ Power
 ☒ Average Irradiance
 ☒ Temperature AmbientAverage
 ☒ Temperature Module Average
 ☒ Performance Ratio

Below the list of variable names and to the right are three icons. The first provides the variables used in the graph in a table format. (Use your browsers back arrow to return to the overview page.) The second switches the display to a bar chart. The third enables downloads of the displayed data in .xlsx format.

In the upper right corner of the tab is an icon which enables you to print the image or save it in a variety of formats. PNG, JPEG, SVG, and PDF are supported.

Figure 89. Power, Irradiance, Ambient and Module Temperature: Columns style control



The Inverter Status Site Tab

The Inverter Status tab features graphical representation similar to that employed at the Fleet Level for each plant. However, rather than displaying the number of inverters (plus associated panels) operating at each of the four states, the individual inverters are each represented by their own icon. As before, Green indicates normal operation, yellow and orange for warning and significant fault, respectively, and red for critical fault.

Information about the events at each array is given in the Site Events tab, below.

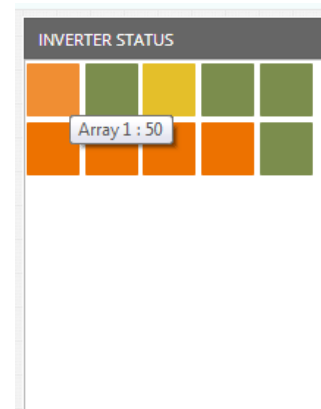
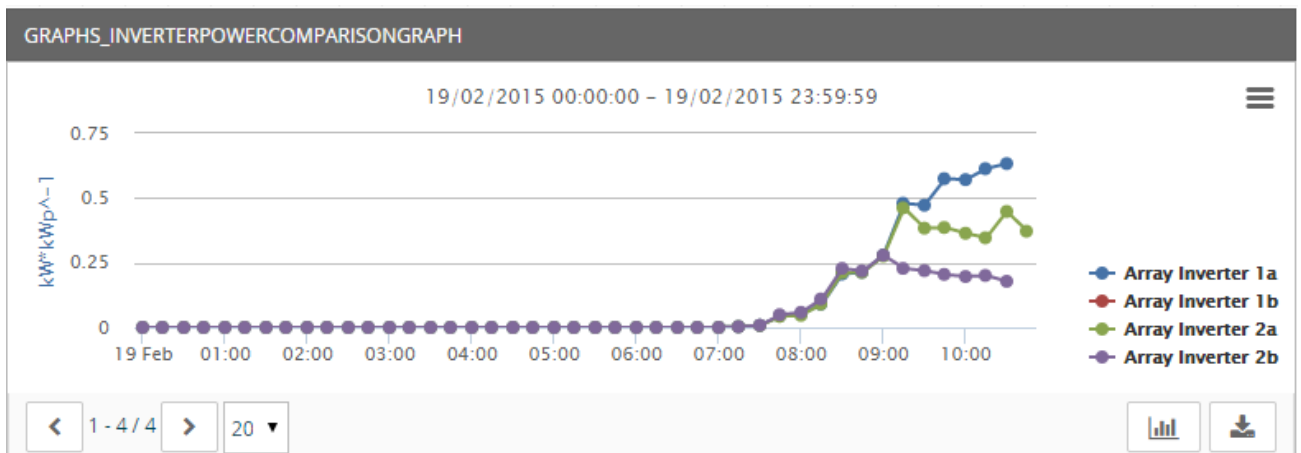


Figure 90. Inverter Status control

The Inverter Power Comparison Graph Tab

The second horizontal layer of information on the page begins with the Inverter power comparison graph. This is also a key graphical display tab that allows quick comparison of power output for multiple arrays at a plant. The time controls at the top of the page affect this graph, and the zoom-to-area feature is available here.

Figure 91. Inverter Power Performance graph control



For plants with a large number of inverters, two additional controls located in the bottom-left corner of the graph are useful. The right control offers the ability to define the maximum number of inverters displayed at once, while the left one offers the ability to scroll between these groups if multiples exist.

Figure 92. Displaying Capabilities Filtering Controls



This tab offers selectable representation of each inverter. The legend of existing arrays is displayed in the right side of the control. Click on the individual legends to add or exclude them from the graph. The same export, data storage and display alternation functionality exists as is presented elsewhere on the Overview page. (Refer to the discussion of the Main Performance graphs for details.)

Figure 93. Removing inverter performance curves

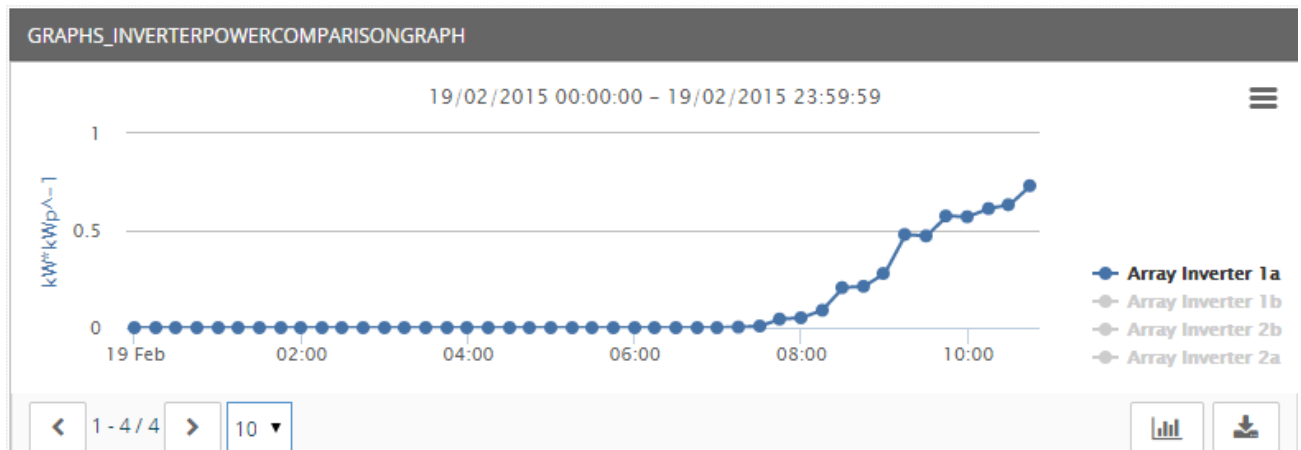
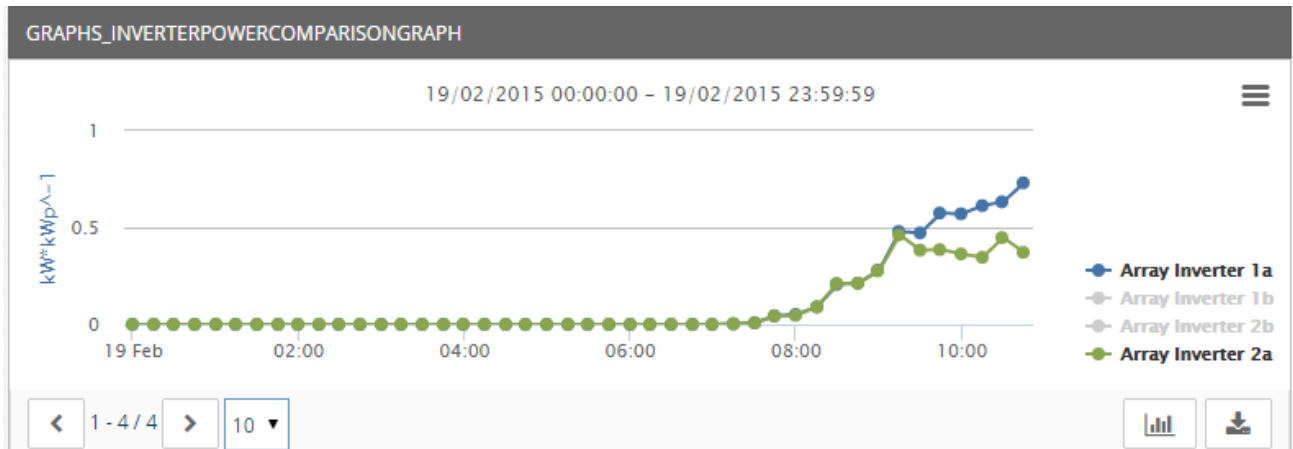


Figure 94. Comparative results for inverter units



The Site Events Tab

The “Site Events” tab control is also a primary control presenting a list concentrating every event detected at the PV Plant.

SITE EVENTS				
SEVERITY	DESCRIPTION	STATE	DATE	SOURCE
	status	Abnormal,Current Abnormal		
60	String current abnormal status	String Current Abnormal,Current Abnormal	19/02/2015 08:22:10	Panel Group 2b-5-15
60	String current abnormal status	String Current Abnormal,Current Abnormal	19/02/2015 08:22:10	Panel Group 2b-5-13
60	String current abnormal status	String Current Abnormal,Current	19/02/2015 08:22:06	Panel Group 1a-2-21

Figure 95. Site
Events List
display control

Each event record contains informational elements that offer the ability to further understand the event:

- Severity (the criticality of the event according to a predefined ranking method)
- Description (a short descriptive text)
- State (the observed condition that triggers the event)
- Date (the date-time stamp of the event)
- Source (the device name that produces the event, according to the name pattern displayed within the “Plant Browser” page)

On the right-bottom corner of the tab area an event filtering control offers you the ability to enter a text string in order to perform a search the list of events. Each column header can drive a forward or reverse sort. Once a condition is no longer happening, it is removed from the displayed list. However it remains in the record of the site and can be accessed through the features described in the Reporting section.

Figure 96. Using the event filtering control

SITE EVENTS				
SEVERITY	DESCRIPTION	STATE	DATE	SOURCE
60	String current abnormal status	String Current Abnormal,Current Abnormal	19/02/2015 12:14:00	Panel Group 1a-7-17
60	String current abnormal status	String Current Abnormal,Current Abnormal	19/02/2015 12:14:00	Panel Group 1a-7-16
60	String current abnormal status	String Current Abnormal,Current Abnormal	19/02/2015 12:14:00	Panel Group 1a-7-14

< 1 - 100 / 132 > 100 ▼ current 🔍

The two additional controls located in the bottom-left corner of the tab offer extra filtering capabilities. These are helpful in the case of numerous event records. The right control offers the ability to define the maximum number of events displayed in the window. The left one offers the ability to scroll between these groups.

Figure 97. Displaying Capabilities Filtering Controls

<	1 - 100 / 132	>	100 ▼
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The Site Arrays Tab

The “Site Arrays” tab control provides a summary view of the production from major elements of a plant. Depending on the topology of the plant, this may include the level of arrays immediately beneath in inverter.

Figure 98.
Site Arrays
List display
control

SITE ARRAYS				
ARRAY	DC POWER	ENERGY TODAY	POWER	PR
Array Inverter 1a	489.78	75.35	136.70	87.87
Array Inverter 1b	486.48	566.50	0.00	99.40
Array Inverter 2a	491.15	75.48	135.42	87.78
Array Inverter 2b	486.48	82.02	135.84	89.96

< 1 - 4 / 4 > 100 ▼ Filter 🔍

Displayed parameters include:

- DC Power (rated power per string, in kWp)
- Energy Today (energy produced until now, in kWh)
- Power (present power production, in kW)
- PR (performance ratio, in percent)

As with the other graphs on this page, each column header can drive a forward or reverse sort, a search function is available, and the number of entries displayed per page can be controlled.

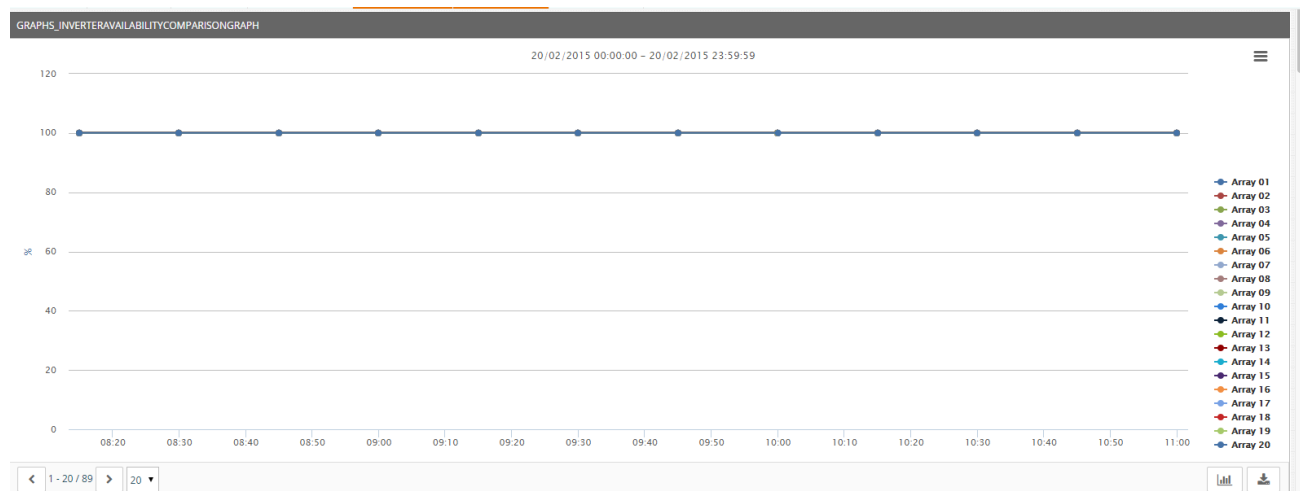
Optional Graphs

The graphs described below are not included in the default layout of the Site Overview page, but may be included at your option. The means of adding these graphs is described at the end of this section.

The Inverter Availability Tab

The “Inverter Availability” tab is an optional graph that graphically demonstrates the inverter availability percentage value (%) for every existing inverter device. The same controls apply on this graph as any other, namely the ability to add or remove devices, change to a bar graph, view data, export data, print or save the graph, and limit the number of items displayed per page. Refer to the description of the key graphs included in the default configuration for a complete explanation of these features.

Figure 99. Inverter Availability graph



The Power vs Irradiance Tab

The “Power vs Irradiance” tab is an optional graph that graphically illustrates the association between overall produced power and monitored irradiance. Here you are able to specify the graph involved magnitudes easily by clicking on the down side placed legend control. Additionally you can choose between line and bar illustration format, download graph in several predefined icon format types, throughout the available command set.

Figure 100. Power vs Irradiance line graph

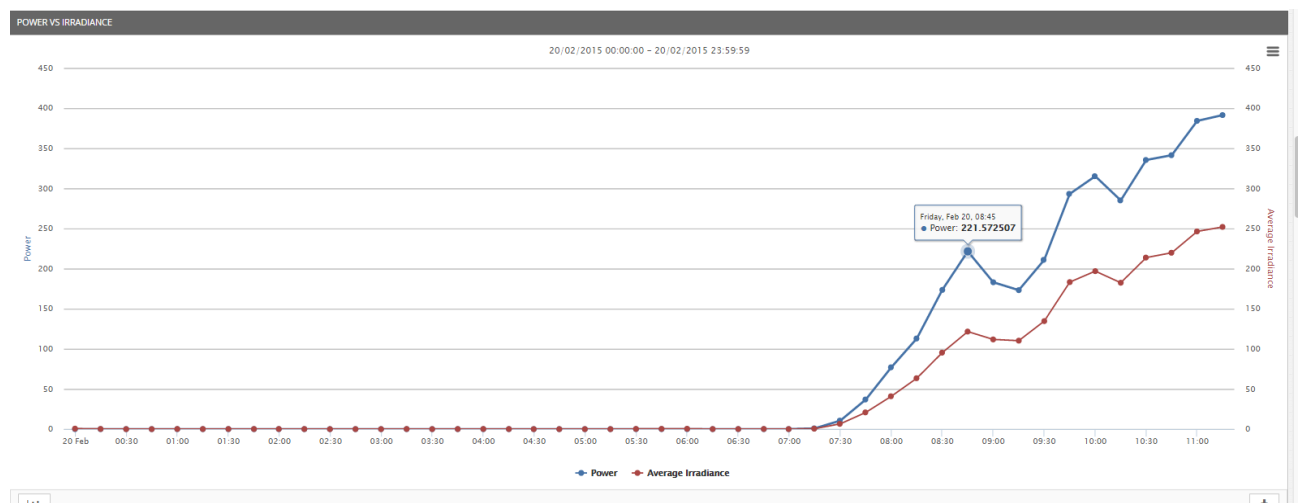


Figure 101. Power vs Irradiance bar graph

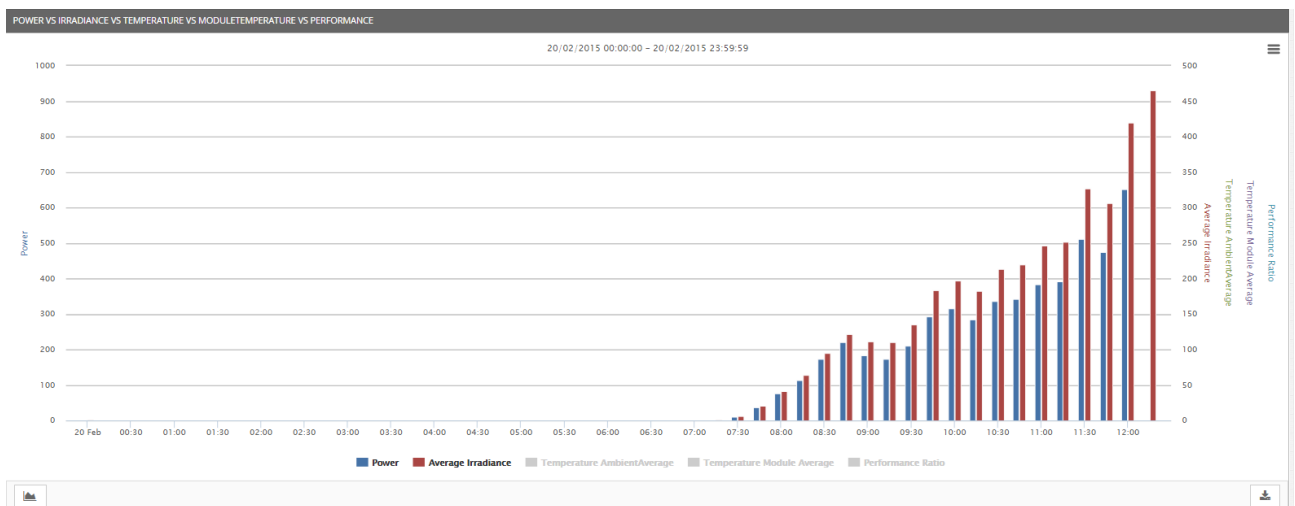
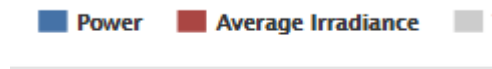


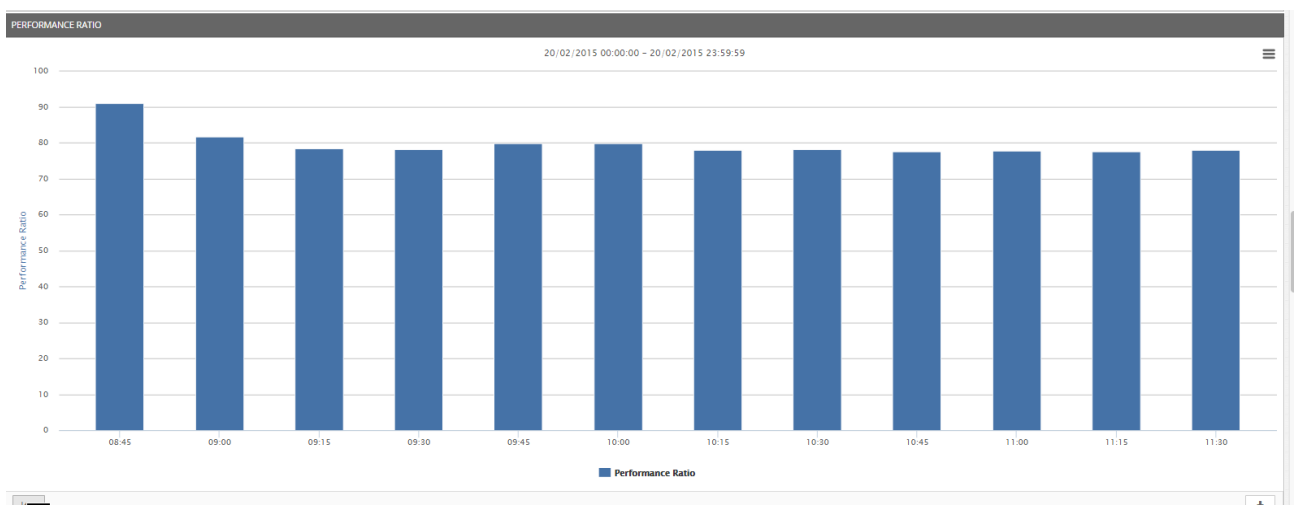
Figure102. The legend selection control



The Performance Ratio Tab

The “Performance Ratio” tab is an optional graph that illustrates the calculated performance ratio for the whole PV Plant site in percentage levels (%). You can switch between line and bar format, download graph in several predefined icon format types, throughout the available command set.

Figure 103. Performance Ratio bar graph



Configuring Overview Display Options

This version of the interface introduces a set of advanced displaying configuration capabilities, allowing customization of the default display. The configuration button placed at the bottom-right side of the header panel enables this mode. This control is automatically activated when you enter the “Overview” page. Upon selecting this mode, the Overview page is transformed to a formatting template.

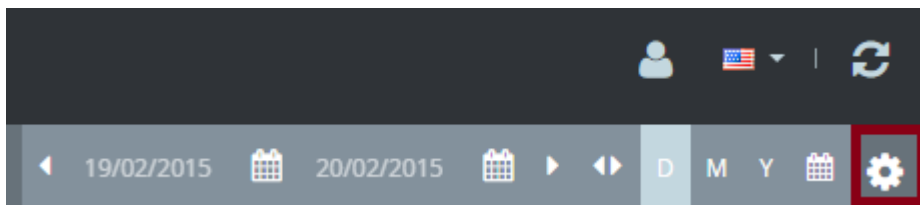
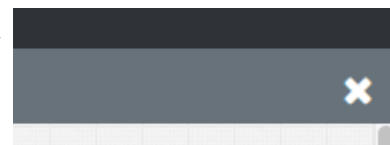


Figure 104. Header Panel Configuration Command Control

At this time, the header panel is replaced with a close button. This allows you to save your changes, exit configuration mode, and return back to the Overview page.

Figure 105. Header Panel Configuration Command Control



During configuration mode every tab is displayed with a similar top-right close button meaning that the User is able to remove the corresponding tab control from the dashboard layout.

Figure 106. The respective Overview page configuration mode style

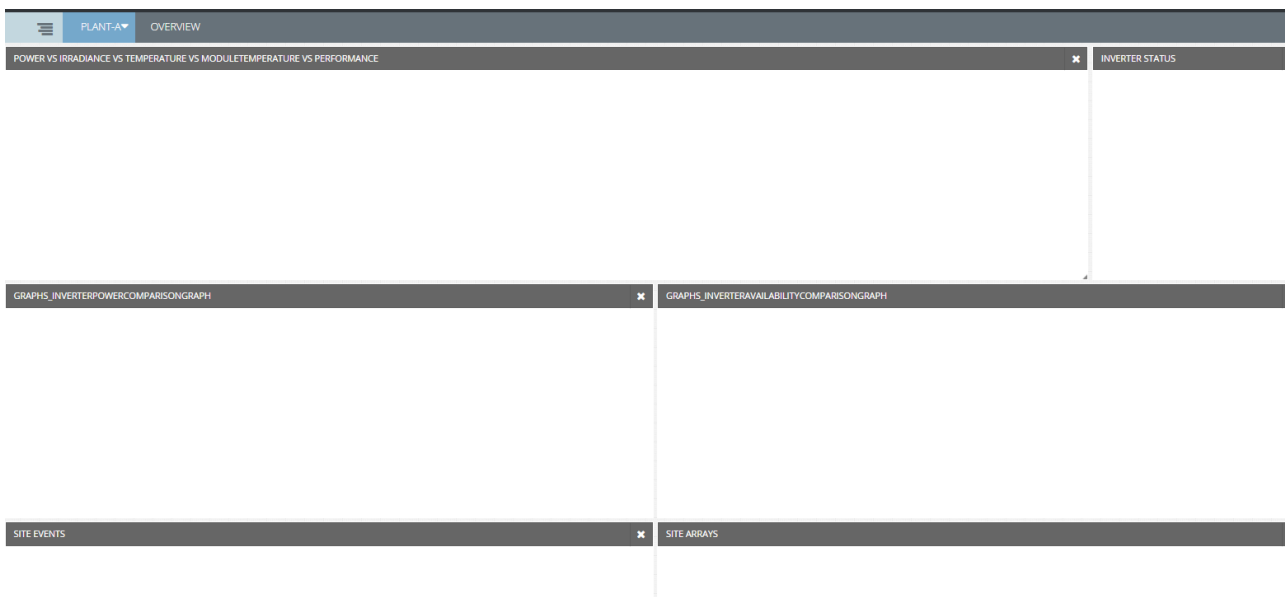
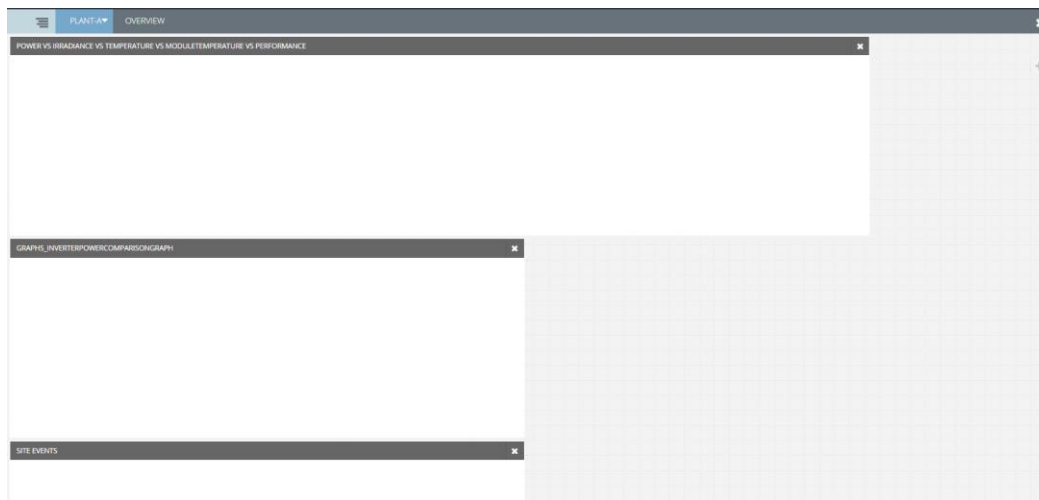


Figure 107. Removing existing tab controls



You are able to perform common resize and drag and drop activities to reconfigure the page to your liking..

Additional tabs may be attached with the a (+) command at the top-right side of the page. To exit the mode of adding new tabs, click the (-) button.

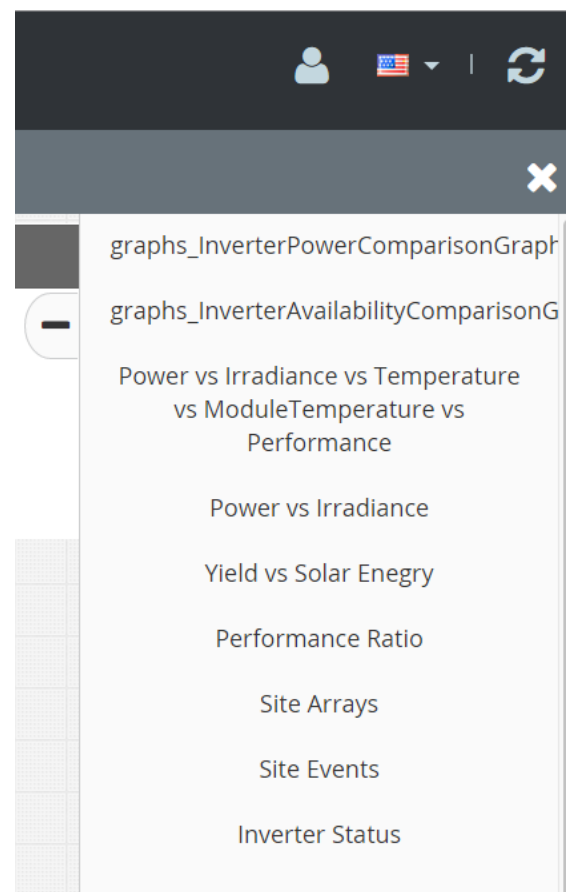
Figure 108.

The tab insertion command button



Figure 109.

The tab insertion command button



Site Level: Browser

The site browser offers a plethora of supplementary features that serve the provision of advanced monitoring techniques. Throughout this page you have the ability to get a fast and accurate knowledge regarding essential PV Plant site installation parameters, segments' operational statuses, ongoing or past events of diverse criticality. Furthermore, you are given the opportunity to select the events of interest for further inspection and the definition of monitoring parameter groups for further analysis. Finally you are also enabled to create notifications assigned to specific events according to particular severity thresholds

The Browser menu command belongs to the Site navigation panel subsection and is only displayed after the proper selection of a PV Plant site from the Fleet Management list or Fleet Map.

Clicking Browser menu command transits you to the Browser page. At the same time header control is reformed. In addition to the time controls seen earlier, the choice of PV Plant can be changed with the pull down feature next to the chosen plant name.

The Browser page is consists of a horizontal frame with numerical informative frame similar to that in the Site Overview and a compact content area. This content area is composed of a browsing tree tab at the left, which can be expanded to show the hierarchy of equipment at the site. The first tab to the right of the tree shows General Data & Parameters. The second displays Events/State/Subscriptions. A third tab, at the bottom of the page is the Preview/Export tab. This integrates the most significant functional aspects of the Browser page.



Figure 110.
Using the
“Browse”
command
button

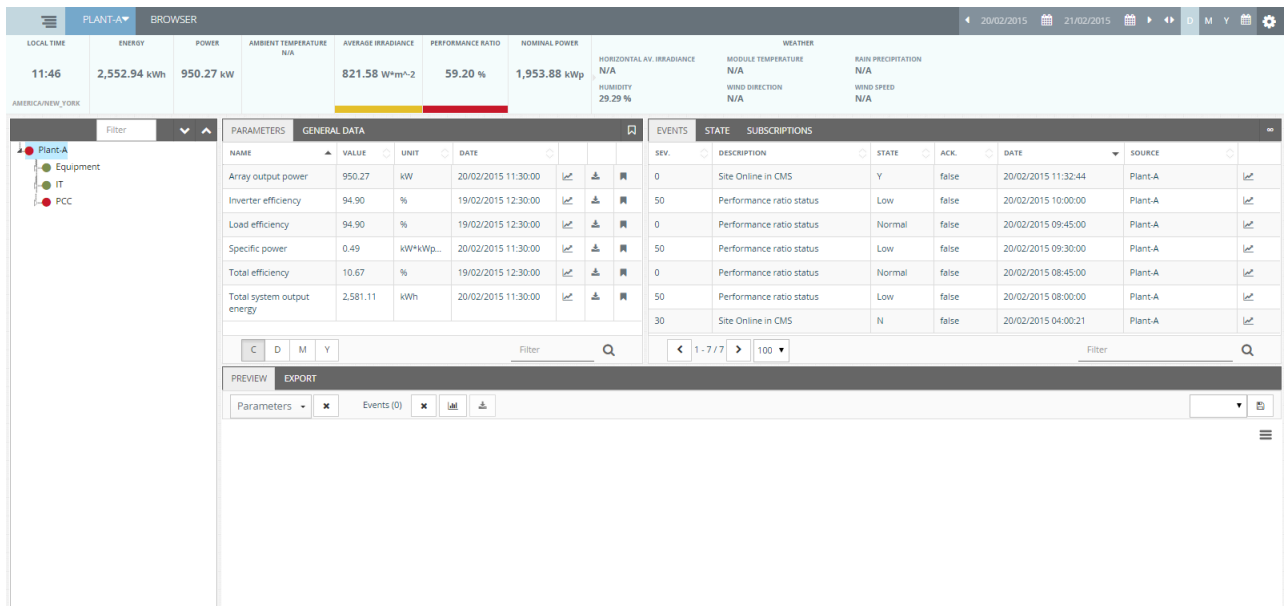
Figure 111. The “Browser” page corresponding header panel



Figure 112. Top side placed Site Numerical information Parameters

LOCAL TIME	ENERGY	POWER	AMBIENT TEMPERATURE	AVERAGE IRRADIANCE	PERFORMANCE RATIO N/A	NOMINAL POWER	WEATHER		
18:39	25,924.02 kwh	0.00 kw	10.29 c	0.00 W/m ²		7,996.80 kwp	HORIZONTAL AV. IRRADIANCE N/A	MODULE TEMPERATURE 12.99 C	RAIN PRECIPITATION N/A
EUROPEROME							HUMIDITY N/A	WIND DIRECTION N/A	WIND SPEED N/A

Figure 113. The Browser page



The Tree Tab

The tree tab is an abstracted representation of the PV Plant site. The tree structure mimics the hierarchy of equipment at the PV Plant. Individual nodes may be expanded to show further dependent nodes. The color indicators adjacent to the node names carry the highest alert level of any dependent node. Thus, for example, if the top node of the plant tree is green, you can be certain that all nodes of the plant are functioning normally. Alternately, to find the source of any event, simply expand the color-indicated nodes until the lowest affected level is reached.

The top of the tree tab is a Filter window. Entering a text string will expose and highlight all matching nodes within the hierarchical structure.

The information shown in the two tabs to the right of the Tree tab is always displayed in context of the node selected at left.

Figure 114.
The Tree
Tab

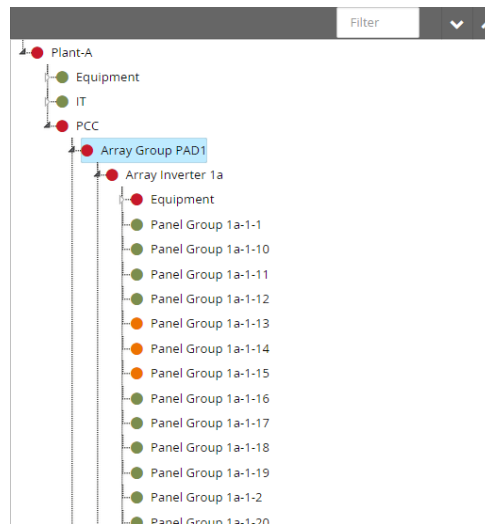
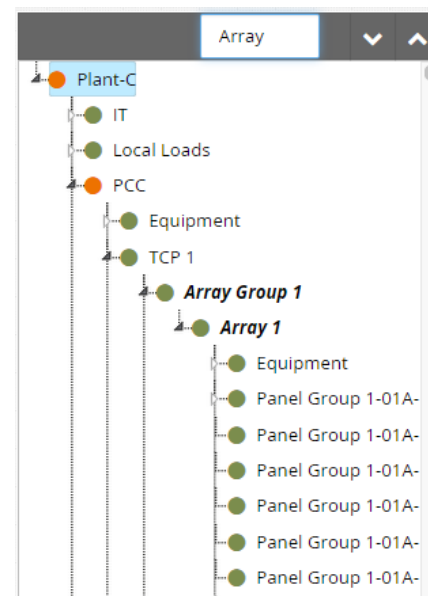


Figure 115.
Using the
Filtering
control



The User has also the ability to expand or collapse the illustrated nodes by using the appropriate up or down arrows respectively.

Figure 116. Using the Expand Control

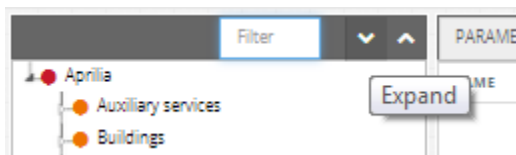
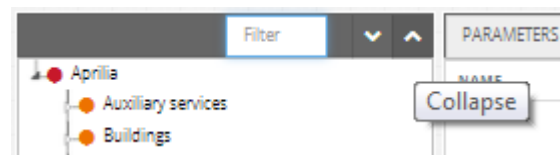


Figure 117. Using the Collapse Control



The Parameters Tab

This tab contains some of the most important information provided by the Inaccess platform. This is a responsive tab refreshed according to the User Tree node selection. The parameters available for each Tree node are presented in a table with the following columns:

- Name, which contains the name of each parameter
- Value, which contains the latest reported value of each parameter
- Unit, which contains the engineering unit of each parameter
- Date, which contains the date and time of the last value of each parameter,
- Preview, which contains a small screen icon if this feature is supported for the parameter
- Export, which contains an icon used to add a parameter to the Export tab. The use of the Export feature and the *scratch pad* shall be further explained below.
- Bookmark, which defines or removes a bookmark from the record

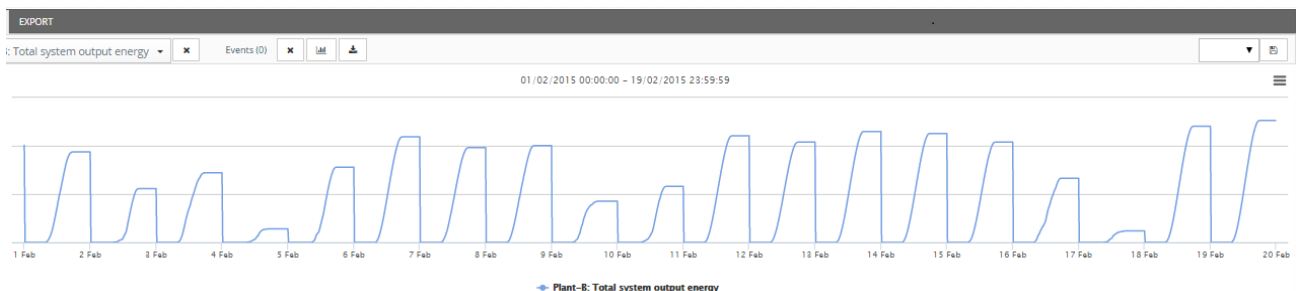
Figure 118. Selecting the parameters Tab

The preview option gives you the ability to attach the selected parameter from the Parameters tab to another tab of the content area - the Preview tab. Preview tab provides the graphical illustration of attached parameters for a selected time-period (for more details please consult the Preview Tab section).

PARAMETERS		GENERAL DATA					
NAME	VALUE	UNIT	DATE				
Array active output power	450.60	kW	20/02/2015 12:00:00				
Array availability	100.00	%	20/02/2015 11:45:00				
Array capture losses	0.35	h*d^-1	20/02/2015 11:45:00				
Array L1-L2 output voltage	502.33	Vac	20/02/2015 12:00:00				
Array L2-L3 output voltage	503.23	Vac	20/02/2015 12:00:00				
Array L3-L1 output	504.26	Vac	20/02/2015 12:00:00				

Figure 119. Selecting a parameter preview

PARAMETERS		GENERAL DATA					
NAME	VALUE	UNIT	DATE				
Total efficiency	10.25	%	23/02/2015 17:00:00				
Inverter efficiency	92.50	%	23/02/2015 17:00:00				
Specific power	0.00	kW*kWp^-1	24/02/2015 05:00:00				
Load efficiency	0.00	%	24/02/2015 05:00:00				
Array output power	0.00	kW	24/02/2015 05:00:00				
Total system output energy	0.00	kWh	24/02/2015 05:00:00				

Figure 120. The corresponding Preview Tab

The export option offers the ability to define a set of parameters for export. By clicking the export icon of each record, the parameter is automatically attached to the export tab in the bottom of the content area. A popup dialog box confirms the addition of the parameter to the export list. You are allowed to select any number of parameters to export. To complete the export procedure, the list of selected parameters becomes a "Scratch Pad." This functionality is further explained within the Export Tab section.

Figure 121. Selecting a parameter export

PARAMETERS GENERAL DATA				
NAME	VALUE	UNIT	DATE	
Total efficiency	10.25	%	23/02/2015 17:00:00	⌵ ⬆ ⌴
Inverter efficiency	92.50	%	23/02/2015 17:00:00	⌵ ⬆ ⌴
Specific power	0.00	kW/kWp~1	24/02/2015 05:15:00	⌵ ⬆ ⌴
Load efficiency	0.00	%	24/02/2015 05:15:00	⌵ ⬆ ⌴
Array output power	0.00	kW	24/02/2015 05:15:00	⌵ ⬆ ⌴
Total system output energy	0.00	kWh	24/02/2015 05:15:00	⌵ ⬆ ⌴
Export				
C D M Y Filter				

Figure 122. The corresponding Export Tab

			⌵	⌴
NAME				
Total system output energy			⌴	

The bookmarking functionality is another method that allows the definition of a set of critical parameters of interest that you wish to monitor in a single tab. First, select the parameters of interest by clicking on the corresponding bookmark icon displayed in the last column of the parameter record. After selecting the complete set of desired bookmarked parameters, select the “Bookmark Show Control” icon in the upper right corner of the tab. The Parameters tab is automatically reformed to contain only the selected parameters. Select the Bookmark Show Control icon again to undo the filter.

Figure 123. Bookmark a parameter

PARAMETERS GENERAL DATA									
NAME	VALUE	UNIT	DATE						SEV.
Total system input energy	0.10	kWh	24/02/2015 05:30:00	⌵ ⬆ ⌴	⌴	⌴	⌴	⌴	50
Total efficiency	10.25	%	23/02/2015 17:00:00	⌵ ⬆ ⌴	⌴	⌴	⌴	⌴	0
Total plant availability	100.00	%	23/02/2015 17:45:00	⌵ ⬆ ⌴	⌴	⌴	⌴	⌴	30
Total Irradiance	0.00	W/m ²	24/02/2015 05:30:00	⌵ ⬆ ⌴	⌴	⌴	⌴	⌴	0
Total system output energy	0.00	kWh	24/02/2015 05:30:00	⌵ ⬆ ⌴	⌴	⌴	⌴	⌴	30
Total revenue	0.00	USD	24/02/2015 05:30:00	⌵ ⬆ ⌴	⌴	⌴	⌴	⌴	0
Bookmark									
0									
30									
0									
C D M Y Total									

Figure 124. Activating Bookmarked Parameters Display

EVENTS STATE SUBSCRIPTIONS			
Show only bookmarked parameters			
⌵ ⬆ ⌴	50	Performance ratio statu	

Figure125. The corresponding Parameters Tab

PARAMETERS		GENERAL DATA			
NAME	VALUE	UNIT	DATE		
Total revenue	0.00	USD	24/02/2015 05:30:00		

C D M Y

Total

The contents of the parameter table can be classified in an ascending or descending order by clicking on every column header and using the up/down arrow. Parameters may be filtered the same as elsewhere in the application by using the filter window in the bottom right of the tab. The content of the control are automatically refreshed and only the records matching the specified filtering pattern are displayed. To remove the filter, clear the text string from the search window.

Figure 126. Applying filtering options

PARAMETERS		GENERAL DATA					
NAME	VALUE	UNIT	DATE				
Total efficiency	10.25	%	23/02/2015 17:00:00				
Inverter efficiency	92.50	%	23/02/2015 17:00:00				
Load efficiency	0.00	%	24/02/2015 04:00:00				

Apart from the filtering control, four other buttons offer the ability to define the time scale used in this tab.

- The “C” button (“C” stands for current)
- The “D” button (“D” stands for daily)
- The “M” button (“M” stands for monthly)
- The “Y” button (“Y” stands for yearly)

The General Data Tab

This tab contains general data related also to the selected Tree node. Each Tree node contains a different set of general data, depending on its type (plant, array group, array etc.). You can add or delete a parameter value within the selected Tree node. Use the + icon at the the upper right of the tab.

Figure 127. Selecting the General Data Tab

PARAMETERS		GENERAL DATA	+
NAME	VALUE		
Name	Array West		
Status	60		
Number of PV panel groups	8		
Multi-string PV array	No		
Blocking diodes	Yes		
Number of PV panels per string	18		
By-pass diodes	Yes		
Concentrator	No		

The Events Tab

This is also an informative tab embedded within the content area of Browser page.

The event table is consisted by the following columns:

- Severity, which contains the severity related to the state that each state-change event led to,
- Description, which contains the name of each state-set,
- State, which contains the state that the state-change event led to,
- Acknowledgement status, which contains the acknowledgement status (true or false) of the state-change event,
- Date, which contains the date/time of the state-change event,
- Source, which contains the name of the tree node which was the source of the state-change event. Though this column may seem redundant, its use shall be further explained below,
- A source link (visible only in the case of recursive event search),
- The Preview icon that gives User the ability to attach this event on the Preview tab.

This presents, upon your first access, the history of the state-change events of the selected Tree node in chronological order for the current day (i.e. from 00:00 until now). The time span configurator header panel functionality is supported for events as well, providing you a quick and efficient tool to modify the time frame for the presented events without entering the configuration panel. The events presented are in context of the node selected in the Plant Tree.

Be advised that choosing to view events spanning a wide time-frame may result in long waiting times due to the large size of the data that have to be retrieved.

Figure 128. Selecting the Events Tab

EVENTS STATE SUBSCRIPTIONS							
SEV.	DESCRIPTION	STATE	ACK.	DATE	SOURCE		
0	Availability level status	Normal	false	22/02/2015 07:45:00	Plant-B		
100	Availability level status	Low	false	22/02/2015 07:30:00	Plant-B		
0	Site Online in CMS	Y	false	20/02/2015 21:17:03	Plant-B		
30	Site Online in CMS	N	false	20/02/2015 19:45:01	Plant-B		
0	Performance ratio status	Normal	false	20/02/2015 08:15:00	Plant-B		
0	Availability level status	Normal	false	20/02/2015 07:45:00	Plant-B		

< 1 - 99 / 99 > 100 ▼ Filter 

Apart for the state-change event table this tab also contains:

- A record page scrolling control,
- An events per page definition control,
- An event filtering edit control.

As with other similar tabs, the events are presented in pages of specified length. You can select the number of events per page and can browse through the event pages using the green arrows at the bottom and left of the event table.

Due to numerous event records and in order to minimize the time and effort need for the detection of a specific event of interest, the search window capability is particularly useful here. Select the column which you desire to search and enter the desired text string in the filter window.

Additional filtering abilities are offered throughout the “Recursive” tick box, which retrieves the events from all Tree nodes below the selected one. Whenever the configuration is altered by the user, the “Refresh” button has to be pressed.

Placing the mouse over severity, description, state, acknowledgement status, date and source column headers exposes the the current sorting pattern used for the classification of records (up/down arrows).

- Up arrow reflects an ascending sort: Selecting this option sorts the contents of the table in an ascending order using the data of the selected column as the key.
- Down arrow reflects a descending sort: Selecting this option sorts the contents of the table in a descending order using the data of the selected column as the key.

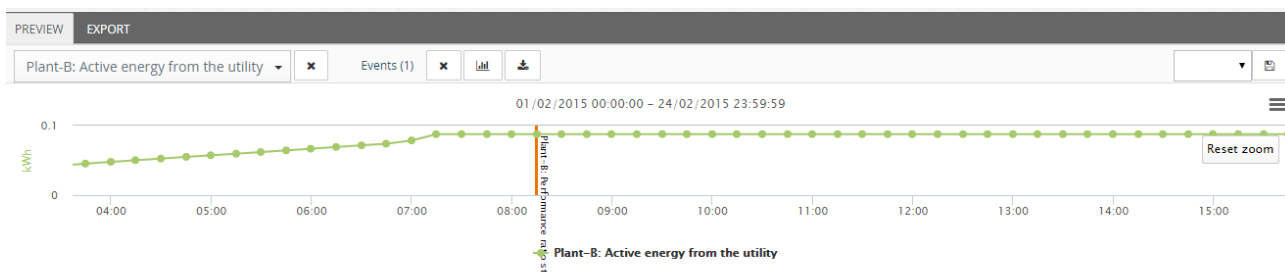
Figure129. Using the sorting functionality

EVENTS STATE SUBSCRIPTIONS	
SEV.	DESCRIPTION
30	Site Online in CMS



Every event record is associated with a graph icon to its right. This is the Preview button which whenever is pressed, the selected event is plotted on a Preview tab as a scatter plot, one axis presenting the event severity and the other one the event time. If you move the mouse pointer over a specific event, a pop-up appears with further event details.

Figure 130. Previewing an event



The State Tab

This tab contains important information regarding the state of the corresponding tree node. The state-sets available for each tree node are presented in a table with the following columns:

- Severity, which contains the severity related to the current state for each state-set,
- Description, which contains the name of each state-set,,
- State, which contains the current state for each state-set,
- Acknowledgement status, which contains the acknowledgement status (true or false) of the current state for each state-set,
- Date, which contains the date/time when the tree node entered the current state,
- Source, which contains critical information regarding the source of this state-set.

Figure 131. Selecting the State Tab

EVENTS STATE SUBSCRIPTIONS						
SEV.	DESCRIPTION	STATE	ACK.	DATE	SOURCE	
0	Performance ratio status	Normal	false	20/02/2015 08:15:00	Array West	
0	DC Voltage Status	Normal	false	20/02/2015 07:00:11	Array West	
0	AC Voltage Status	Normal	false	19/02/2015 18:02:43	Array West	
0	Availability level status	Normal	false	28/01/2015 08:00:00	Array West	
0	Frequency Status	Frequency normal	false	30/05/2014 15:07:59	Array West	

< 1 - 5 / 5 > 100 ▼

Filter



The time span configurator header panel functionality is also supported for state, providing you with a quick and efficient way to modify the time frame for the presented events without entering the configuration panel.

NOTICE: Choosing to view states spanning a wide time-frame may result in long waiting times due to the large size of the data that have to be retrieved.

Apart for the state table this tab also contains:

- A record page scrolling control,
- An states per page definition control,
- A state filtering edits control.

As elsewhere in the application, the states are presented in pages of specified length. You can select the number of states per page and can browse through the pages using the green arrows at the bottom and left of the state table.

To quickly find a specific state, use the search window capability. Select the column you desire to search and enter the desired text string in the filter window.

Placing the mouse over every column headers is leading to the display of the current sorting pattern used for the classification of records (up/down arrows).

- Up arrow reflects sort ascending capability: Selecting this option sorts the contents of the table in an ascending order using the data of the selected column as the key.
- Down arrow reflects a descending sort: Selecting this option sorts the contents of the table in a descending order using the data of the selected column as the key.

The Subscriptions Tab

Figure 132. Selecting the Subscriptions Tab

EVENTS		STATE		SUBSCRIPTIONS		∞		+	
SOURCE		TYPE		CONDITION		THRESHOLD			
Aprilia		EMAIL		ABOVE		90			
Filter									

This tab presents the table of the active notification subscriptions associated with your account for the selected site, together with the necessary controls for them.

Upon first access, the table presents the notification subscriptions that are active for the selected tree node. The table has the following columns:



- Source, which contains the tree node for which the specific notification has been configured. This is useful when all notifications for the specific plant are presented (see below).
- Type, which contains the notification method to be used for the specific notification subscription (e-mail or SMS)
- Condition, which contains the condition to be applied on the threshold (above or below)
- Threshold, which contains the severity threshold above or below which a notification shall be issued
- Trash Can, the icon which enables deletion of the subscription.

NOTICE: Be aware that notifications are issued based on the severity of unacknowledged events, so that if a new (and unacknowledged event) appears, you are notified, regardless of the severity that was imposed by acknowledged events.

You can sort your notification subscriptions by clicking on the arrows facing downwards or upwards inside the header of each column:

- Sort ascending: Selecting this option sorts the contents of the table in an ascending order using the data of the selected column as the key.
- Sort descending: Selecting this option sorts the contents of the table in a descending order using the data of the selected column as the key.

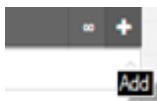
The contents of the notifications table can be filtered in the same way as presented in the previous sections. First, select the column on which the search criterion shall apply. Next, enter the desired search string in the search window.

Adding and Deleting a Notification Subscription

To add a new notification subscription, click on the green “plus” button at the top-right corner of the tab. This opens a new form with pre-defined source information, depending on the associated tree node. To begin the subscription, complete the form with the following mandatory fields:

- Notification type: available options are e-mail and SMS
- Severity threshold: the numerical level of event, above or below which, a notification shall be generated for the specific subscription
- Condition: the condition to be applied on the threshold (above or below).

Figure 133. Using the “plus” (+) button



You have the ability to terminate the process before completion and at any time before this by clicking on the “x” button at the end of the row. To save the subscription, click the “J” (save) button.

Figure 134. Define the new notification subscription type

EVENTS	STATE	SUBSCRIPTIONS			∞	+
SOURCE	TYPE	CONDITION	THRESHOLD			
Array West	EMAIL ▼ EMAIL SMS	ABOVE ▼	10 ▼	✓	✕	
Filter 						

Figure 135. Define the new notification subscription triggering condition

EVENTS	STATE	SUBSCRIPTIONS					∞
SOURCE	TYPE	CONDITION	THRESHOLD				
Array West	EMAIL ▼	ABOVE ▼ ABOVE BELOW	10 ▼	✓	✕		
Filter 							

Figure 136. Define the new notification subscription triggering threshold

EVENTS STATE SUBSCRIPTIONS				∞	+
SOURCE	TYPE	CONDITION	THRESHOLD		
Array West	EMAIL	ABOVE	10	✓	✗
			10		
			20		
			30		
			40		
			50		
			60		
			70		
			80		
			90		
			100		

Filter

Figure 137. Saving the notification subscription request

				∞	+
	CONDITION	THRESHOLD			
	ABOVE	10	✓	✗	

Filter

Figure 138. Successful creation of a new notification subscription

EVENTS STATE SUBSCRIPTIONS					∞	+
SOURCE	TYPE	CONDITION	THRESHOLD			
Aprilia	EMAIL	ABOVE	90			
Filter 						

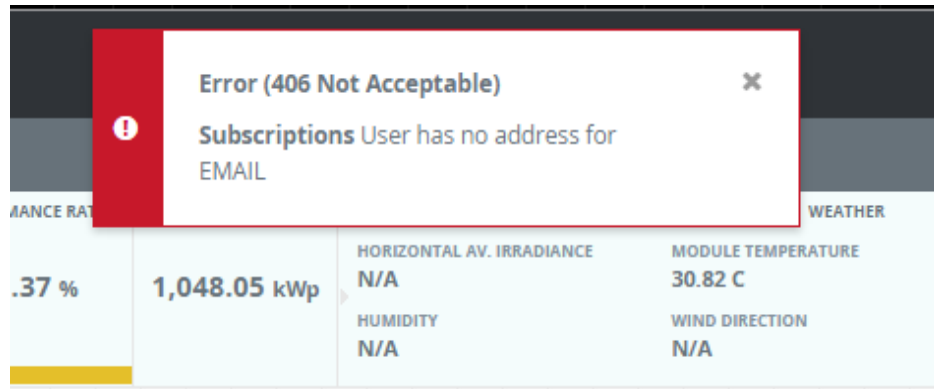
You can deactivate and remove a stored notification subscription by selecting the appropriate record and clicking on the trash can icon displayed at the right-side of the record. The system will request your confirmation depicting two buttons “x” and “/”.

Figure 139. Deleting an existing notification subscription

EVENTS STATE SUBSCRIPTIONS					∞	+
SOURCE	TYPE	CONDITION	THRESHOLD			
Aprilia	EMAIL	ABOVE	90	 		
Filter 						

Important Notice: Please verify that your user profile has your correct email account or mobile phone number in order to activate email or SMS notification subscriptions. If this information is missing, you will not be able to subscribe to an alert and will receive the following error message.

Figure 140. Subscription Creation Error



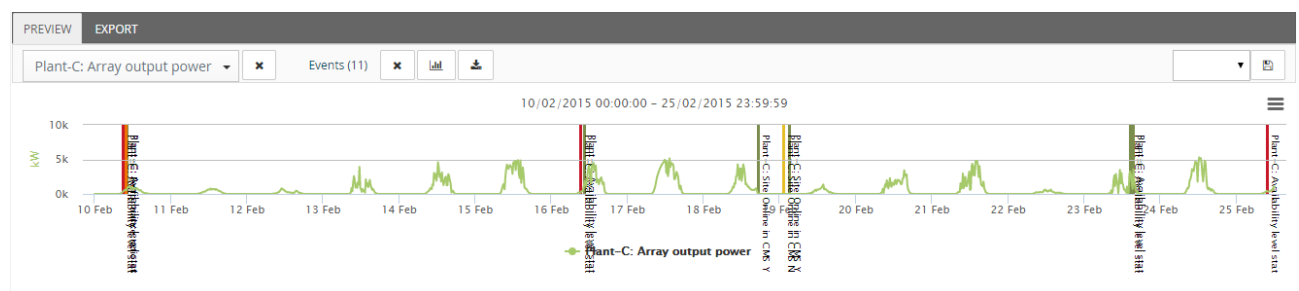
The Preview Tab

This tab is responsible for the graphical representation of the user-selected parameters and events chosen elsewhere on this page. The graph can display any data input combination and present it over the period driven by the same time controls used elsewhere in the application. The graphical representation of parameter variation and event occurrences is a strong feature allowing the ability to detect an abnormality and the source of it. As such, the user can extract significant conclusions about the site operational status. The Preview tab is a powerful compound monitoring tool that enhances the default system provided monitoring abilities by offering an open functional philosophy. The user is free to decide the combination that will provide the best graphical insight into the health and performance of the PV plant.

The parameters and events available for use in the Preview Tab are those identified from within the Parameters or Events tabs, by clicking on the chart icon in their respective rows. Note that although the Parameters and events will change with the node selections in the Tree Table, the list collected in the drop-down menus of the Preview Tab will retain the various selections made at different nodes.

It is possible to save a collection of Parameters and Events for future use, and to recall stored sets of elements. Note that these are stored with association to your login, but is portable to other sites.

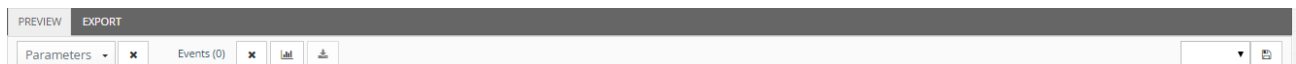
Figure 141. The Preview Tab



The Preview tab toolbar offers several commands at the top of the window:

- Parameter pull-down Listing, Select or Deselect the Parameters collected from above
- Events pull-down Listing, Select or Deselect the Events collected from above
- Show Log, used to view the numerical data used in the graph
- Lines icon, to switch between Line and Bar graph style
- Export, to download the parameters and events displayed in a csv file
- Stored Graph Pull-down, to display an already stored graph
- Save, to store current graph for future use

Figure 142. Preview tab toolbar controls



Attaching Parameters

In order to attach a new parameter on the Preview Tab, you must first select the relevant Tree node. The Parameters tab updates to show the monitoring parameters associated with that node. You can then use the Preview icon button (graph symbol to the right of the date column) to attach the parameter to the Preview tab. The Preview tab will be automatically reformed, drawing the variation of the selected parameters over the time period specified by the time controls in the second bar of the page.

Attaching Events

In order to attach a new event on the Preview Tab, first select the relevant Tree node. The Events tab updates to show the events associated with that node. You can then use the Preview icon button (graph symbol to the right of the date column) to attach the parameter to the Preview tab. The Preview tab will be automatically reformed, drawing the occurrence of this specific event. The time controls in the second bar of the page are active in this window.

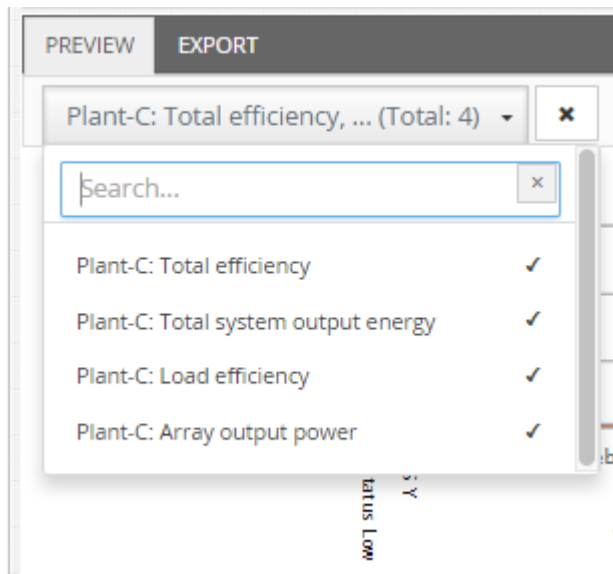
Filtering Attached Parameters

Whenever a new parameter or event type is added, the corresponding Parameter or Event drop-down menu is refreshed to display the cumulative number of participating preview items.

Figure 143. Updated Parameter and Event filtering controls



Select or deselect an item from these lists for inclusion in the Preview graph. Each menu contains a full list of selected parameters or events respectively. Figure 144. Using the parameters filtering dialog



The graph will update to respond to your choices. Items may be selected and deselected repeatedly.

The parameters may also be included or excluded from the preview graph by clicking on them in the legend, immediately below the graph.

Figure 145. The displayed parameter legend

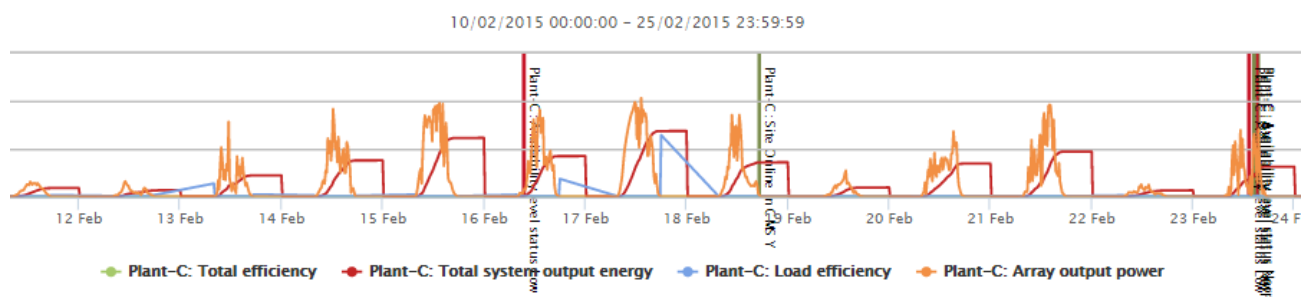
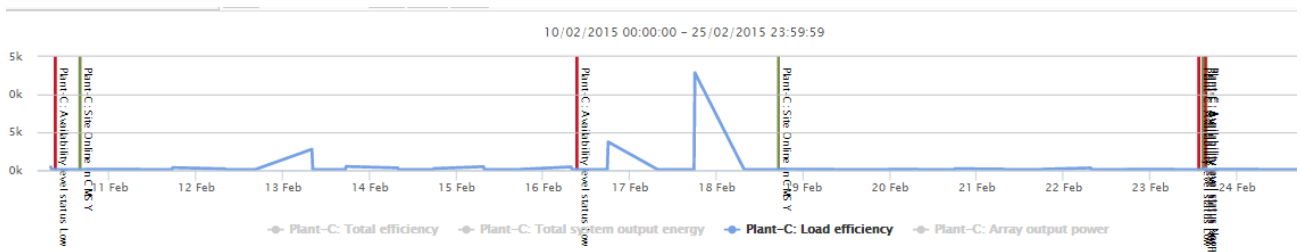


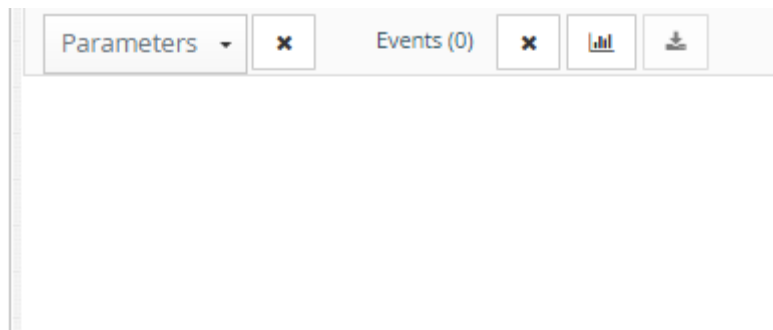
Figure 146. Clicking on legend items



Permanently Detaching Preview Items

To permanently clear the collection of Parameters or Events in the preview menu, click the “X” to the right of each drop down menu.

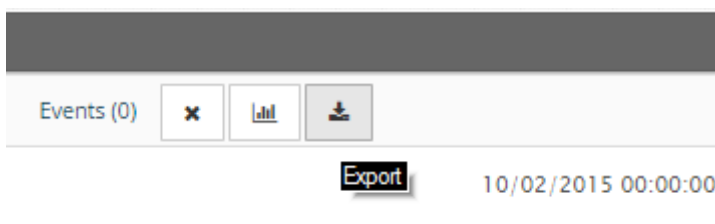
Figure 147. Clearing Parameters and Events



Exporting graph data parameters

After selecting the desired Parameters and Events, and specifying the time period for the graphical presentation in the Preview window, it is possible to export the participating data in a csv file. Clicking on the “Export” icon on the top-center bar of the Preview tab will do that.

Figure 148. Using the Export button



Graph image manipulation

Once the Preview graphs is as you like it, you may print or export the graph image through the Chart Context Menu at the top right of the graph.

Figure 149. Activating the Chart context menu

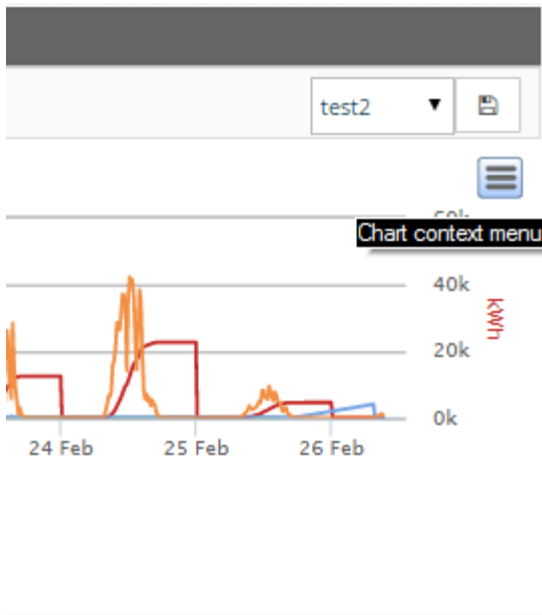
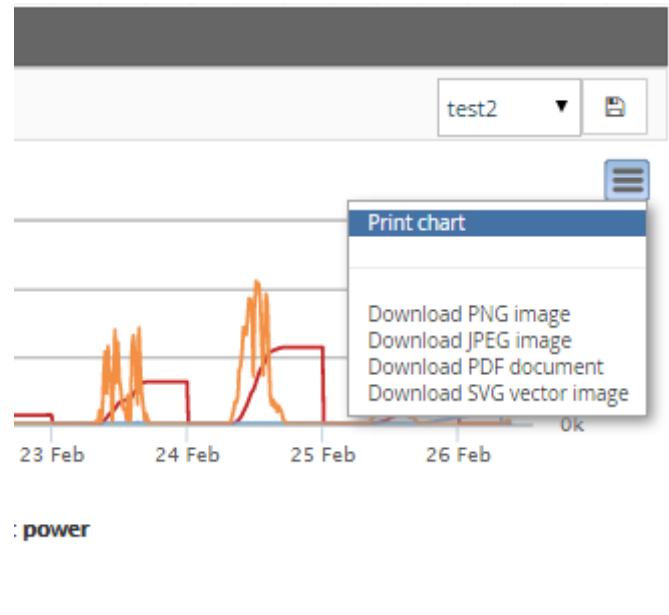


Figure 150. Selecting the preferable image exporting type



Storing Parameter and/or Event Association Pattern

Also in the Preview tab is the ability to store a parameter and event group for future use. Clicking the “Save” button will prompt you to enter a graph pattern name. This saved pattern will be able for reuse in different PV Plants and for different time periods. This saves the process of redefining the participating parameters and events.

If you select a saved parameter and event group for use in the Preview tab, the parameter and event lists will be updated.

Selecting graph displaying options

The preview tab offers the choice between “Line” or “Bar” style graphs. Select the button in the top bar showing either the graph or bar icon.

Figure 151. The Lines Style

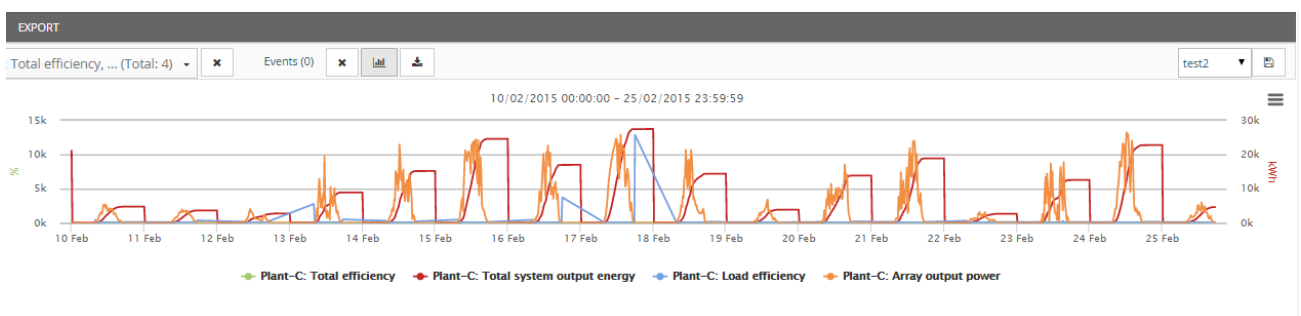
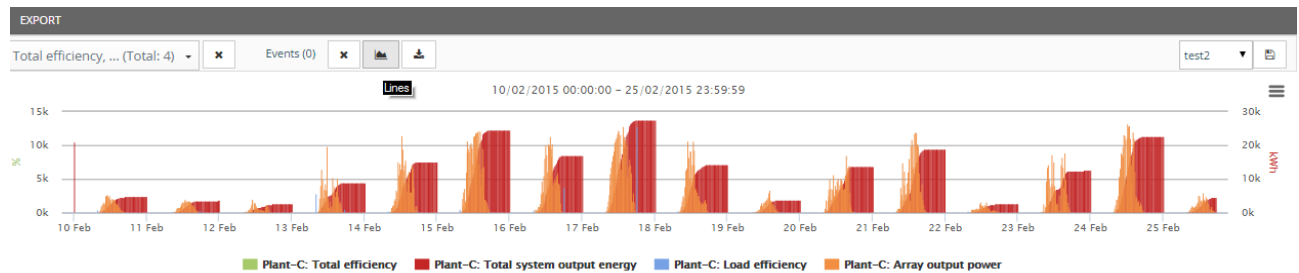


Figure 152. The Bars Style



The Export Tab

The custom measurement parameters used in the Preview tab can be saved and reused, as described above, and are also savable through the “Scratch Pad” feature of the Export Tab in the Preview window. (This concept does not apply to Events, just parameters.)

Creating a new Scratch Pad Template

In order to create this custom Scratch Pad, first attach each desired parameter by selecting the appropriate Tree node under which it appears, then select the “Export” icon button adjacent to the parameter in the Parameters window. A pop up message confirms the successful addition of the parameter to the Export List.

Figure 153. Selecting a parameter export

PARAMETERS				
GENERAL DATA				
NAME	VALUE	UNIT	DATE	
Total efficiency	10.25	%	23/02/2015 17:00:00	⌵ ⌴ ⌶
Inverter efficiency	92.50	%	23/02/2015 17:00:00	⌵ ⌴ ⌶
Specific power	0.00	kW/kWh~1	24/02/2015 05:15:00	⌵ ⌴ ⌶
Load efficiency	0.00	%	24/02/2015 05:15:00	⌵ ⌴ ⌶
Array output power	0.00	kW	24/02/2015 05:15:00	⌵ ⌴ ⌶
Total system output energy	0.00	kWh	24/02/2015 05:15:00	⌵ ⌴ ⌶
Export				
C D M Y Filter Q				

Figure 154. The corresponding Export Tab

		▼	📄
◇	NAME	◇	
	Total system output energy		📄

Each inserted record is consisted of the:

- SOURCE, reflecting the parameters' segment of reference,
- NAME, reflecting the parameter name, identical to the parameter system name.

To remove any parameter from the list, click on the trash can icon button displayed on the right side of the record in the parameter list.

Upon the completion of your export list, export the template by clicking on the “Save” icon at the top right of the tab. You will be prompted to name the Scratch Pad template.

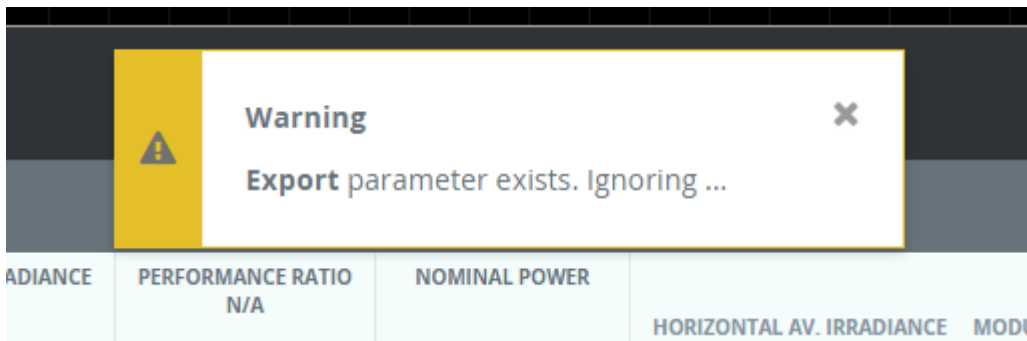
Figure 155. Saving the new Scratch Pad template

New scratch pad

Name

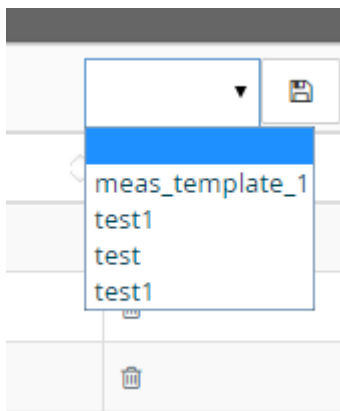
Save

An individual parameter may be inserted only once. In case of duplication, the following error is displayed.

Figure 156. Automatic Prohibition of parameter overlapping

Using an Existing Scratch Pad Template

From within the Preview window, you have the ability to open an existing Scratch Pad template by clicking on the drop down window next to the save icon. This displays a list of existing Scratch Pad template records. Once opened, the template can be modified by insert additional parameters or removing existing ones. Remember to resave and edited template.

Figure 157. Selecting an Existing Scratch Pad Template

Selecting Scratch Pad Templates from the Report Subscription page

Once a Scratch Pad template is saved, it may be used also as the basis for Report Subscriptions. While some commonly used templates exist, this method allows complete flexibility in creating Report Subscriptions focused on areas of concern.

General: Web client application messages to the user

Application messages to the user appear in a popup window and the user is prompted for an action. The table below summarizes the messages and their cause.

Message	Meaning
<u>Messages caused by user actions</u>	
<i>All events updated successfully!</i>	This message informs you that all events were acknowledged successfully
<i>Not all the events were updated.</i>	This message informs you that the acknowledgement of some events has failed
<i>Setting(s) well updated</i>	This message informs you that all settings were updated successfully
<i>One setting failed to be updated, please try again</i>	This message informs you that the update of some settings has failed
<i>No data retrieved</i>	This message appears when you have to configured (using the “Graph” sub-tab configuration) a time-span in which no data are available.

Message	Meaning
<u>Messages caused by the internals of the application</u>	
<i>Cache expired: reload the data from first page</i>	In the “Event handling” page, the set of data retrieved and shown have expired, so the application has to reload the event list and return to the first page of the event paged table
<i>Application will restart: You have been logged out. Please relog!</i>	This message appears when the authentication of the user has expired. The user has to relog
<i>Application will restart: <error text></i>	This message appears when the application has experienced an internal error. You have to contact Inaccess and provide the error text.